# Variational and Optimization-based Approaches for Event-Based Vision Applications

## From unobserved modes to well-posed estimators

Daniel Blauvelt

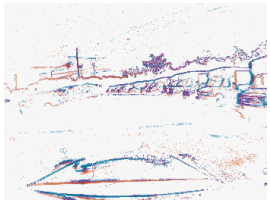
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Dissertation Defense ▪ Fall 2026

# What can changes tell us about a scene?



**Reference image**

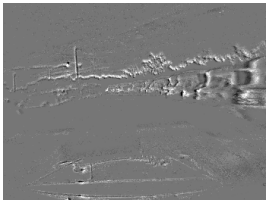


**Accumulated events**

50 ms sample

Orange +; blue −; purple: both.

→



**Our reconstruction**

What do the observations determine, and what does the model supply?

**Takeaway.** The observations constrain changes; the model supplies the remaining choices.

Dissertation: Chapter 2, saved MVSEC [35] outdoor_day1 reconstruction example.

# The route through the talk

| | |
|---|---|
| **Sensor and data** | What does an event camera report? |
| **Reconstruction** | How can changes recover an image over time? |
| **Optical flow** | How can local measurements determine motion? |
| **Stereo depth** | What changes when two cameras estimate depth? |
| **Conclusions** | What have we learned, and what comes next? |

Reconstruction is the main focus, followed by the flow theorem and a brief stereo summary.

**Takeaway.** The common question: what do the observations determine, and what does the model supply?

Dissertation: Table 5.1.

# Where this work fits

## Image reconstruction

Rebecq et al. [12]: recurrent network, events to video.

Zhang, Yezzi & Gallego [6]: optical flow supplies a linear image inverse problem; classical or learned priors.

Antil–Sayre [2]: events and frames; bilevel estimation of an event-based blur-kernel parameter.

**This work:** per-pixel temporal inversion from event increments; kernel representations, activity-based penalties, and fixed-weight guarantees.

## Optical flow

Horn–Schunck [16]: brightness constraint and spatial smoothness.

Benosman et al. [50]: timestamp geometry.

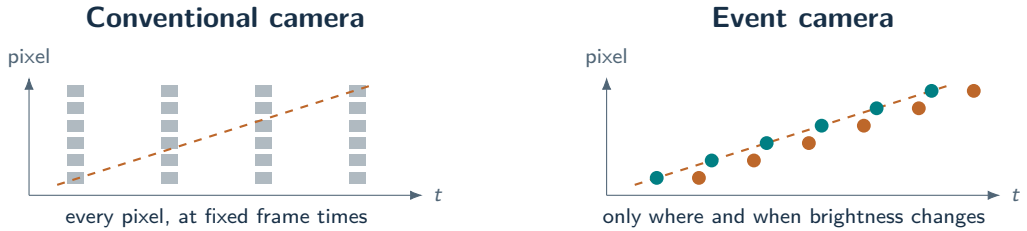
Weickert–Schnörr [57]: convex-regularizer framework and well-posedness theory.

**This work:** three event representations in one quadratic theory, with explicit identifiability and stability assumptions.

**Takeaway.** The contribution is the event-derived models and their precise guarantees: what the data determine and what regularization supplies.

Dissertation: Chapter 2 and Section 3.2; temporal publication: Antil, Blauvelt, and Sayre [17].

# Why an event camera? Each pixel reports its own changes



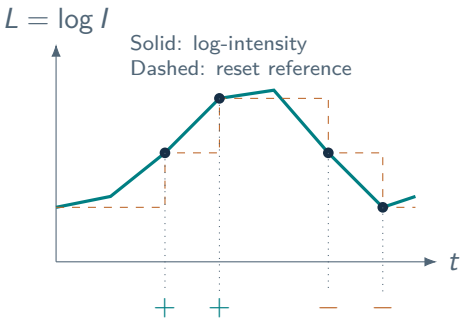

Dashed line: the leading edge of a moving bright bar. Teal: brightening (+); amber: darkening (−).

- Microsecond timestamps, about 140 dB dynamic range versus 60 dB, low power, little motion blur [1].
- Under steady illumination, motion makes events trace boundaries; illumination changes can also trigger events.

**Takeaway.** The sensor trades absolute brightness for precise timing of changes; each estimator supplies what the changes leave free.

Dissertation: Chapter 2, neuromorphic camera overview; Gallego et al. [1]. Diagram is schematic.

# An event camera: each pixel reports threshold crossings




An event is a tuple

$$e_k = (\mathbf{x}_k, s_k, p_k), \qquad p_k = \pm 1.$$

For the ideal threshold model,

$$L(\mathbf{x}_k, s_k) - L(\mathbf{x}_k, s_k^-) = p_k c_{\mathrm{th}}.$$

The preceding reset $s_k^-$ is **local to that pixel**.

**Takeaway.** Events record changes. Adding a constant to a pixel's log-intensity leaves its events unchanged.

Dissertation: Eq. 2.1; Eq. 3.3; ideal threshold model [20]. Diagram is schematic.

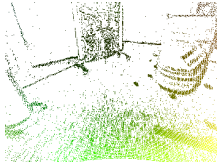
# From events to images, motion, and depth



**Image reconstruction**

Ch. 2

Grayscale: reconstructed value.

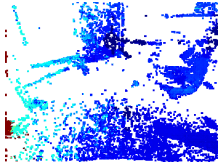


**Optical-flow estimate**

Ch. 3; event-active pixels



hue: direction
brightness: size
(full at ≈ 14 px)



**Stereo-depth estimate**

Ch. 4

depth (jet), near to far



0.5 8 m

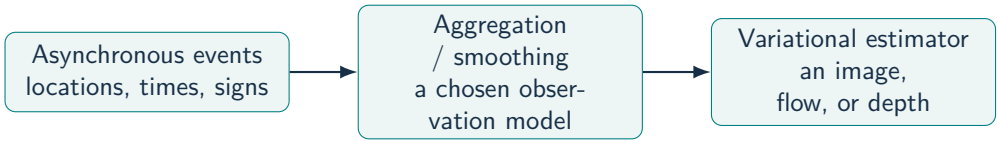
MVSEC [35] indoor_flying2, drone-mounted event cameras, near 47.6 s; stereo depth uses both.

**Takeaway.** Each problem leaves a different choice: a brightness baseline, motion along an edge, and which pixels correspond.

Dissertation: Methods: Chapter 2, Chapter 3, and Chapter 4; MVSEC [35].

# A point measure needs a representation before optimization

$$\mathcal{E} = \sum_k p_k \, \delta_{(\mathbf{x}_k, s_k)}.$$




- Temporal kernels yield integrated changes at reconstruction times.
- Time surfaces and voxel grids yield coefficients for motion constraints.
- Stereo accumulation yields paired temporal event-feature observations.

**Takeaway.** The representation supplies usable coefficients and introduces modeling choices.

Dissertation: Eq. 2.2–Eq. 2.4; Section 3.3; Section 4.3.3.

# Start with one pixel: integrated changes on a temporal chain

Choose a unit-mass temporal kernel $\phi_{s_k}$:

$$e(t) = \sum_{k \text{ at this pixel}} p_k \phi_{s_k}(t).$$

$$y_i = \int_{t_i}^{t_{i+1}} e(t)\, dt.$$

The reconstruction times $t_i$ are chosen by the algorithm.

$$\mathbf{A} = \begin{pmatrix} -1 & 1 & & \\ & -1 & 1 & \\ & & \ddots & \ddots \end{pmatrix}.$$

$$\mathbf{A}\mathbf{f} \approx \mathbf{y}.$$

$$\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}.$$

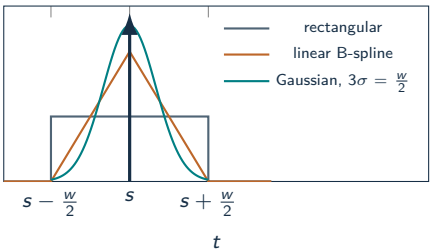
Events determine increments, leaving a constant free.

**In scene terms:** each pixel's log-brightness history is rebuilt on its own from how much it rose or fell in each interval. The model chooses the reference level; a pixel without events stays at that level, whatever its brightness.

**Takeaway.** Kernel width controls temporal averaging; a baseline requires an additional choice.

Dissertation: Eq. 2.4, Eq. 2.7, and Lemma 2.2.4.

# Temporal kernels: how far each event's influence spreads




Arrow: delta kernel (pure binning). Common support width $w$; schematic.

Each kernel has unit mass, is a translate, and has compact support:

$$\int_{\mathbb{R}} \phi_s = 1, \qquad \phi_s(t) = \phi_0(t - s).$$

Each event splits its sign $p_k$ across the bins it overlaps:

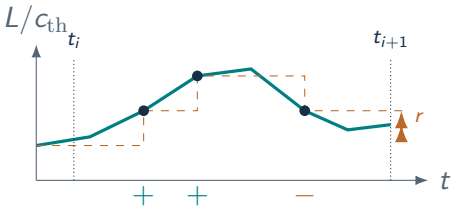
$$y_i = \sum_k p_k \int_{t_i}^{t_{i+1}} \phi_{s_k}(t)\, dt, \quad \sum_i y_i = \sum_k p_k$$

when every support lies inside $[t_0, t_N]$.

**Takeaway.** Kernel width trades suppression of timing noise against blurring of rapid changes. The operator **A** is the same for every kernel.

Dissertation: Proposition 2.2.1, Proposition 2.2.2; Table 2.2.

# From threshold crossings to an approximate observation




Solid: $L/c_{\mathrm{th}}$. Dashed: reference at the last event.

Ideal sensor: each event marks a log-brightness change of about one threshold $c_{\mathrm{th}}$.

$$y_i = \sum_k p_k \int_{t_i}^{t_{i+1}} \phi_{s_k}(t)\, dt \;\approx\; \frac{L(t_{i+1}) - L(t_i)}{c_{\mathrm{th}}}.$$

Exactly, for any times $a < b$ at one pixel:

$$\frac{L(b) - L(a)}{c_{\mathrm{th}}} = y(a,b) - \underbrace{B(a,b)}_{\text{kernel spill at bin edges}} + \underbrace{\frac{1}{c_{\mathrm{th}}} \sum_{a < s_\ell \le b} \varepsilon_\ell}_{\text{overshoot}} + \underbrace{r(b) - r(a)}_{\text{sub-threshold change}}.$$

**Takeaway.** Events approximate scaled brightness changes; the model supplies the level.

Dissertation: Physical connection (p. 24), Eq. 2.29, arbitrary-endpoint identity; threshold model [20]. Schematic diagram.

# A regularized estimate is uniquely selected

For fixed $\mathbf{\Lambda} = \mathrm{diag}(\lambda_i)$ with every $\lambda_i > 0$,

$$\mathbf{f}^* = \arg\min_{\mathbf{f}\in\mathbb{R}^n} \left\{ \frac{1}{2} \|\mathbf{A}\mathbf{f} - \mathbf{y}\|^2 + \frac{1}{2} \|\mathbf{\Lambda}\mathbf{f}\|^2 \right\}.$$

The unique minimizer solves the normal equations:

$$\mathbf{K}_{\mathrm{time}} = \mathbf{A}^\top\mathbf{A} + \mathbf{\Lambda}^\top\mathbf{\Lambda} \succ 0, \qquad \mathbf{K}_{\mathrm{time}}\mathbf{f}^* = \mathbf{A}^\top\mathbf{y}.$$

**Compute each pixel, then assemble frames**

1. Form the pixel's matrix and right-hand side; solve the linear system numerically.
2. Run independent pixel solves in parallel; assemble their temporal profiles into frames.

The selected baseline satisfies $\mathbf{1}^\top\mathbf{\Lambda}^2\mathbf{f}^* = 0$: intervals with larger weights set the reference level.

**Takeaway.** The penalty chooses an unobserved baseline and also changes observable increments.

Dissertation: Eq. 2.8, Theorem 2.2.5; Tikhonov regularization [27]; monocular reconstruction implementation, per-pixel linear solves.

# Why uniqueness and stability hold

Write $\lambda_{\min} = \min_i \lambda_i$. For $\mathbf{z} \neq 0$,

$$\mathbf{z}^\top \mathbf{K}_{\text{time}} \mathbf{z} = \|\mathbf{A}\mathbf{z}\|^2 + \|\mathbf{\Lambda}\mathbf{z}\|^2 \geq \lambda_{\min}^2 \|\mathbf{z}\|^2 > 0.$$

At the same fixed weights,

$$\delta\mathbf{f}^* = \mathbf{K}_{\text{time}}^{-1} \mathbf{A}^\top \delta\mathbf{y}, \qquad \|\delta\mathbf{f}^*\| \leq \frac{\|\mathbf{A}\|_2}{\lambda_{\min}^2} \|\delta\mathbf{y}\|.$$

$$\|\mathbf{A}\|_2 \leq 2, \qquad \kappa_2(\mathbf{K}_{\text{time}}) \leq \frac{4 + \lambda_{\max}^2}{\lambda_{\min}^2}.$$

One spurious or missed event gives $\|\delta\mathbf{y}\| \leq 1$: the bound limits how far one noise event moves a pixel's history.

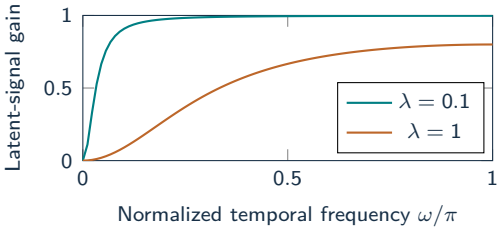
**Takeaway.** Coercivity gives a stable estimate; weakening the penalty weakens these full-space bounds.

Dissertation: Theorem 2.2.5, Proposition 2.2.7.

# What the penalty does to temporal information

For scalar $\mathbf{\Lambda} = \lambda \mathbf{I}$ and noiseless data $\mathbf{y} = \mathbf{A}\mathbf{f}_{\mathrm{true}}$:

$$g_k = 4 \sin^2 \frac{\pi k}{2n}, \qquad H_k = \frac{g_k}{g_k + \lambda^2}, \qquad H_0 = 0.$$




The baseline is removed.

Low temporal modes are attenuated most.

A unique estimate does not restore the missing physical baseline.

In the images: slow brightness drift fades; passing edges remain.

Dissertation: Lemma 4.4.1, Theorem 4.6.21; cosine eigenbasis [81]. Curves: continuous extension.

# How are the temporal weights chosen?

For activity $a_i = |dc_i|$, the selected sigmoid rule is

$$\lambda_i = 10^{-3} + \frac{10}{1 + e^{a_i}}.$$

Here $dc_i$ is the signed event accumulation; $a_i$ is its magnitude, not a count of all events.

| Rule | Sigmoid | Inverse proportional | Exponential |
|---|---|---|---|
| Reported outdoor CN | 29.60 | 81.50 | 100.74 |

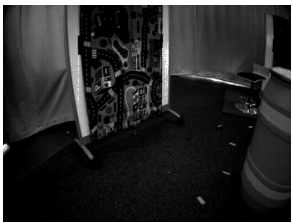
Selected by grid sweeps, balancing image fidelity and conditioning.
Quiet intervals get the largest weights and set the reference level; busy intervals get the smallest and set the worst-case noise bound.

**Takeaway.** Positive weights ensure a unique solve; the stability bound holds with weights fixed.

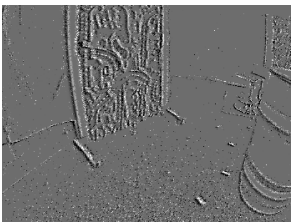
Dissertation: Adaptive temporal weights (p. 38); Table 2.7.

# Temporal evidence: structure without an absolute baseline

MVSEC [35] indoor flying 2



Reference



Event-only reconstruction

MVSEC indoor aggregate results

| Kernel | PSNR | CN |
|---|---|---|
| B-spline | 8.86 | 22.16 |
| Delta | 7.70 | 1375.75 |
| Gaussian | 8.84 | 20.22 |

PSNR compares with a reference frame, so it also scores the absolute brightness that the model selects; the events constrain only changes.

**Takeaway.** Kernel and adaptive-weight choices affect both the reconstruction and the solved system.

Dissertation: Table 2.4; images: MVSEC indoor flying 2, Ch. 2 saved release.

# Relation to Antil–Sayre bilevel reconstruction

**This chapter**
Events only $\longrightarrow$ activity-based $\Lambda(dc)$ $\longrightarrow$ fixed convex temporal solve.

**Antil–Sayre [2]**
Events and intensity frames $\longrightarrow$ outer optimization over $z$, a per-pixel, per-frame parameter of the event-based blur kernel.

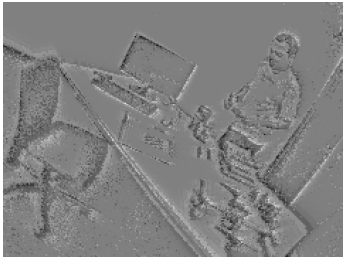
| Sequence | Method | PSNR | CN |
|---|---|---|---|
| Indoor | Bilevel / event-only | 15.37 / 10.85 | 39755 / 25.51 |
| Outdoor | Bilevel / event-only | 8.41 / 8.71 | 39755 / 27.10 |

PSNR: reference-image similarity (dB). CN: condition number of each method's inner linear system.

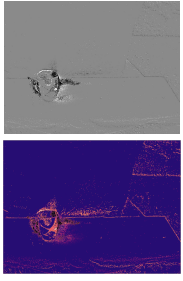
**Takeaway.** Frame information and a different optimization problem change the comparison.

Dissertation: Table 2.8 and Table 2.9. Antil–Sayre [2], Sec. 3, Eqs. (3.1)–(3.3).

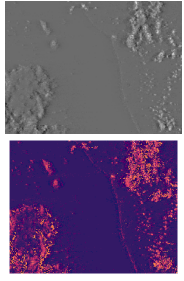
# Real event data: indoor, high-speed, and orbital scenes



**Office**, handheld; ≈ 2.5M events



**Bullet hits a cup**; ≈ 350K events



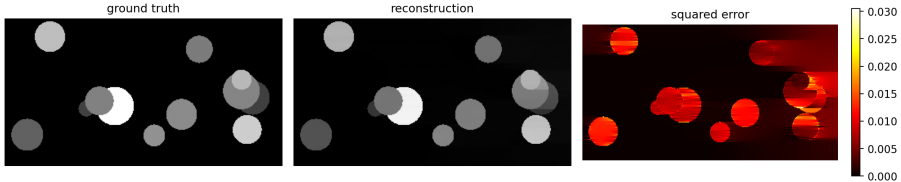
**ISS** (Falcon Neuro); ≈ 2M events

Cup and ISS: grayscale (top) and false color (bottom) of the same reconstructed frame.

**Takeaway.** One per-pixel fixed-weight solve, with sigmoid weights and no intensity frames, recovers structure across very different time scales.

Dissertation: Figure 2.7, Figure 2.8, and Figure 2.9; Section 2.3.3. Data: office [36], cup [37], Falcon Neuro [38].

# Shared motion couples temporal profiles to one scene




Simulated events (v2e [46]); the camera window shifts 1 px per view. Left: scene. Center: shared-scene estimate (smoothed-TV penalty). Right: squared error.

**Chapter 2:** each pixel has its own temporal profile.

**Shared scene:** known motion $P$ maps one static scene $\theta$ into every view.

$$\underbrace{B_{\mathrm{SAR}}}_{\text{scene-to-event operator}} = (A \otimes I)\, P.$$

**Takeaway.** Known motion lets many views constrain one scene. Optical flow asks what happens when the motion is the unknown.

Dissertation: Figure 2.10, Eq. 2.47, Proposition 2.4.1; joint ESAR work [45]; synthetic aperture [98].

# The aperture problem: what a local view reveals

An **aperture** is a restricted observation window: a small view of an edge [66].

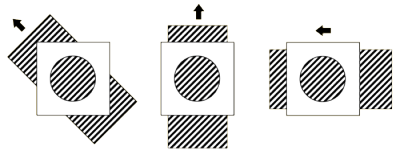
**Through the windows**



The visible stripe motions look the same.

Movie 5 source (GIF)

**Reveal the full patterns**



The patterns translate in different directions.

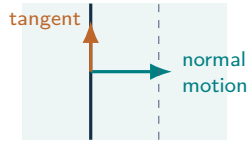
Movie 6 source (GIF)

**Takeaway.** An edge reveals perpendicular (normal) speed; motion along it (tangent) stays free. More observed directions can constrain the full velocity.

Perceptual illustration: Wolfe et al., *Sensation and Perception*, 4th ed., Sinauer © 2015. Activity 8.2, Movies 5–6.
Dissertation: Proposition 3.3.4 and the following aperture discussion.

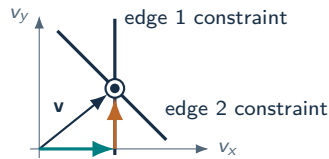
# Motion introduces a new invisible direction

**Image: one moving edge**




A straight edge reveals its normal speed.

**Velocity: what the events allow**




Edge 1 fixes the normal part. If edge 2 shares the motion, it fixes the tangent part edge 1 left free.

**The flow model in scene terms**

- **Unknown:** image-plane velocity at every pixel.
- **Edges:** events give the speed perpendicular to each edge.
- **Neighbors move alike:** supplies motion along edges and where no events occur.
- **Requirement (main theorem):** edges in at least two orientations somewhere, so their lines cross.

**Takeaway.** As with the temporal baseline, the data leave one quantity free; smoothness carries information from other edge orientations to supply it.

Intersection of constraints [Adelson & Movshon 1982]. Dissertation: Proposition 3.3.4; requirement in Theorem 3.6.1.

# Derive the time-surface constraint by following the edge

For an ideal single crossing, define

$$\psi(\mathbf{x}, t) = t - \mathcal{T}(\mathbf{x}), \qquad \psi(\mathbf{x}(t), t) = 0.$$

If the edge translates with constant $\dot{\mathbf{x}} = \mathbf{v}$, then

$$0 = \frac{d}{dt}\psi(\mathbf{x}(t), t) = 1 - \nabla\mathcal{T} \cdot \mathbf{v}.$$

In scene terms: $|\nabla\mathcal{T}|$ is edge slowness, normal speed $1/|\nabla\mathcal{T}|$; motion along the edge leaves $\mathcal{T}$ fixed.

## Model hypotheses

The edge crosses each location once, has nonzero normal speed, and its recorded timestamp equals a differentiable crossing-time map.

**Takeaway.** Interpolation, stale timestamps, and repeated crossings make this an approximate data constraint.

Dissertation: Proposition 3.3.4; Definition 3.3.2; time surfaces [49, 50].

# Voxel projection: transport a represented event-activity field

Smooth and bin events into a spatiotemporal field $V(\mathbf{x}, t)$; project over a window, $a(\mathbf{x}, s) = \int_s^{s+T} V(\mathbf{x}, t)\, dt$.

## Modeling assumption (added to the threshold model)

The activity field moves with the scene: $V_t + \mathbf{v} \cdot \nabla V = 0$, with constant $\mathbf{v}$ over the window.

Integrating over the window then gives

$$a_s + \mathbf{v} \cdot \nabla a = 0, \qquad a_s = V(\mathbf{x}, s+T) - V(\mathbf{x}, s),$$

so the voxel-projection residual is $\nabla a \cdot \mathbf{v} + a_s$.

**Takeaway.** The threshold model supplies the activity field; transporting it with the scene is a separate modeling choice.

Dissertation: Section 3.4.1; derivative-operator qualifications; voxel grid [39].

# The structure tensor pools directional information

At a fixed evaluation time $t_\star$, set

$$\mathbf{J}(\mathbf{x}) = \int W(\mathbf{x}-\mathbf{y})\, \nabla_3 V(\mathbf{y}, t_\star) \nabla_3 V(\mathbf{y}, t_\star)^\top \, d\mathbf{y}, \qquad W \geq 0.$$

Use the local residual energy

$$\mathbf{v}_3^\top \mathbf{J} \mathbf{v}_3, \qquad \mathbf{v}_3 = (v_x, v_y, 1)^\top, \qquad \mathbf{J} = \begin{pmatrix} \widetilde{\mathbf{J}} & \mathbf{j}_t \\ \mathbf{j}_t^\top & J_{tt} \end{pmatrix}.$$

- Each gradient outer product is positive semidefinite and rank at most one.
- Pooling nonparallel spatial gradients can make $\widetilde{\mathbf{J}}$ full rank.

**Takeaway.** Pooling can identify both components locally; the global theorem needs only integrated identifiability.

Dissertation: Section 3.4.2, Proposition 3.4.4; spatiotemporal-surface view [49].

# Three representations, one quadratic variational problem

Rename the flow data matrix $\mathbf{B}$ to distinguish it from temporal $\mathbf{A}$:

$$E_{\mathbf{B},b}(\mathbf{v}) = \frac{1}{2}\int_\Omega (\mathbf{v}^\top \mathbf{B}\mathbf{v} - 2b^\top \mathbf{v}) + \frac{\alpha^2}{2}\int_\Omega |\nabla \mathbf{v}|^2.$$

| Representation | $\mathbf{B}$ | $b$ | Edge measurement from |
|---|---|---|---|
| Time surface | $\nabla\mathcal{T}\nabla\mathcal{T}^\top$ | $\nabla\mathcal{T}$ | crossing times |
| Voxel projection | $\nabla a \nabla a^\top$ | $-a_s\nabla a$ | transported event activity |
| Structure tensor | $\widetilde{\mathbf{J}}$ | $-\mathbf{j}_t$ | gradients pooled over a window |

In scene terms: $\mathbf{B}$ records which motion directions the edges measure, and how strongly; $b$ carries the measured values; the $\alpha$ term asks neighbors to move alike.

**Takeaway.** The representations change the coefficients; the analysis can be proved once.

Dissertation: Theorem 3.6.1 and its three corollaries; data-plus-smoothness form [16].

# The main flow theorem: global information controls constants

## Unified well-posedness theorem

Let $\Omega \subset \mathbb{R}^2$ be bounded, connected, and Lipschitz; $\alpha > 0$. Assume

$$\mathbf{B} \in L^\infty, \quad \mathbf{B} = \mathbf{B}^\top \succeq 0 \text{ a.e.}, \quad b \in L^2, \qquad \boxed{\int_\Omega \mathbf{B}(\mathbf{x})\, d\mathbf{x} \succ 0}.$$

Then $E_{\mathbf{B},b}$ has a unique minimizer in $H^1(\Omega;\mathbb{R}^2)$.

The integrated matrix must identify **every constant velocity direction**.
In scene terms: edges in two or more orientations somewhere, so their constraint lines cross; neighbors supply the motion where $\mathbf{B} = 0$.

**Takeaway.** $\mathbf{B}$ may vanish on open regions; integrated directional information and smoothness still give a unique estimate.

Dissertation: Theorem 3.6.1, Eq. 3.59; classical quadratic-flow theory [57].

# Proof idea: smoothness fixes variation, edges fix the drift

Decompose $\mathbf{v} = \underbrace{\bar{\mathbf{v}}}_{\text{common drift (pan)}} + \underbrace{\mathbf{w}}_{\text{variation across the image}}$, with $\int_\Omega \mathbf{w} = 0$.

**Smoothness controls variation**

$$\|\mathbf{w}\|_{L^2} \le C_P \|\nabla \mathbf{v}\|_{L^2}$$

Poincaré [26]

**Data control constant motion**

$$m = \lambda_{\min}\Big(\int_\Omega \mathbf{B}\, d\mathbf{x}\Big) > 0$$

edges cover every direction

**Together: coercivity**

$$\mathcal{B}(\mathbf{v}, \mathbf{v}) \ge c_0 \|\mathbf{v}\|_{H^1}^2$$

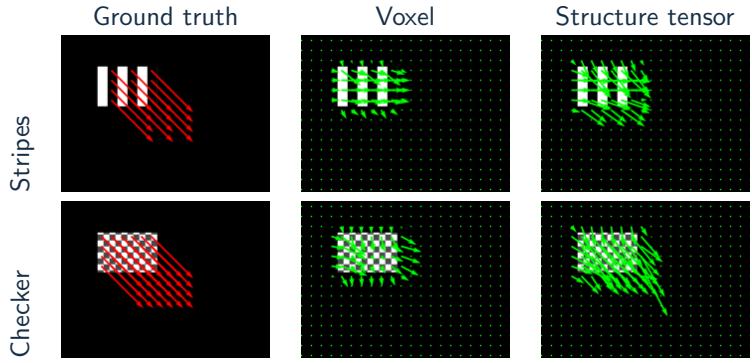
$$c_0 = c_0(\Omega, \alpha, \|\mathbf{B}\|_{L^\infty}, m)$$

In scene terms: smoothness leaves a camera pan free, so the edges must fix it; $c_0$ shrinks with $m$, as for nearly parallel stripes (next slide).

**Takeaway.** The gradient penalty controls nonconstant variation; integrated directional information controls the constant modes. Lax–Milgram [72] then gives the unique minimizer.

Dissertation: proof of Theorem 3.6.1.

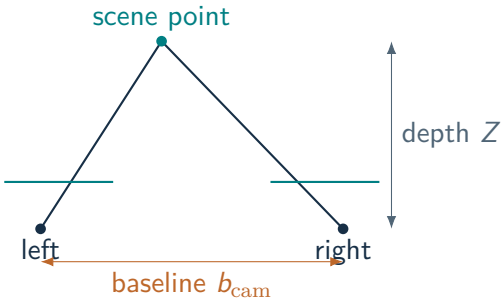
# Known-motion examples make the aperture problem visible




**Takeaway.** Checker texture adds motion directions and reduces the aperture ambiguity; accuracy also depends on the representation and regularization.

Dissertation: Figure 3.4; v2e [46] simulation, same window and arrow scale.

# Two viewpoints turn correspondence into depth




Rectification (row alignment) makes the search horizontal:

$$d = x_L - x_R.$$

$$Z = \frac{f_{\mathrm{cam}}\, b_{\mathrm{cam}}}{d}, \qquad d > 0.$$

Each candidate $d$ pairs a left observation with $\mathbf{y}_R(x - d, y)$.

**In scene terms:** the true match sees the same scene point, so the two observed event signals agree; each candidate is scored by how well one shared explanation fits both views.

**Takeaway.** The correspondence is unknown; solving the latent profiles at a fixed candidate is only the inner problem.

Dissertation: Section 4.5, Eq. 4.11; rectified stereo geometry [80]. Geometry is schematic.

# Stereo on MVSEC: competitive accuracy, more scored points

At each candidate disparity $d$, solve two softly coupled temporal profiles:

$$\min_{\mathbf{f}_L,\mathbf{f}_R} \sum_{c=L,R} \left( \tfrac{1}{2} \left\| \mathbf{W}_c^{1/2} (\mathbf{A}\mathbf{f}_c - \mathbf{y}_c) \right\|^2 + \tfrac{1}{2} \left\| \mathbf{\Lambda} \mathbf{f}_c \right\|^2 \right) + \tfrac{\gamma}{2} \left\| \mathbf{f}_L - \mathbf{f}_R \right\|^2 .$$

**Mathematical result** (fixed $d$, frozen positive weights): Hessian $\succeq \lambda_{\min}^2 \mathbf{I}$; unique inner solve.
**Benchmark:** the full WLS method, with weight updates, spatial cost aggregation, and candidate selection.

| Method | Mean depth error (cm) ↓ FI1 | FI2 | FI3 | FI4 | log RMSE ×100 ↓ | Scored points (M) |
|---|---|---|---|---|---|---|
| **Soft-coupled WLS (ours)** | 73.83 | **58.35** | 70.33 | 59.24 | **44.54** | 2.66 |
| Piatkowska adaptation [78] | **67.96** | 58.75 | **68.88** | **54.63** | 44.72 | 1.20 |
| Xie adaptation [105] | 93.43 | 78.60 | 93.66 | 66.57 | 59.36 | 1.39 |
| Kogler et al. [104] | 123.39 | 99.50 | 125.27 | 78.23 | 80.94 | 0.66 |
| Rogister et al. [103] | 142.17 | 128.20 | 153.32 | 114.80 | 90.32 | 1.01 |

MVSEC indoor flying 1–4, raw outputs. Each method is scored on its own estimates with ground truth in $[0.5, 8]$ m.

**Takeaway.** WLS: best on Flying 2 and log RMSE, within 6 cm elsewhere, over twice the points.

Dissertation: Theorem 4.6.3; Table A.5, Table A.4; MVSEC [35]. Xie and Piatkowska rows are local adaptations.

# What the data constrain and what the models prove

| Problem | What the observations constrain | What the mathematical model establishes |
|---|---|---|
| Temporal reconstruction | Represented brightness increments | Unique regularized history for each fixed positive weight matrix, with perturbation bounds |
| Optical flow | Event-derived directional motion constraints | Unique $H^1$ minimizer under integrated identifiability |
| Stereo depth | Paired temporal event-feature observations | Unique frozen-weight inner solve; the disparity comes from a separate comparison of candidates |

**Takeaway.** Under the stated hypotheses, regularization makes each estimator determinate. Physical accuracy also depends on how well the representation describes the scene.

Dissertation: Theorem 2.2.5, Theorem 3.6.1, Theorem 4.6.3; Table 5.1.

# Research opportunities

## Event-sensor modeling

Quantify the represented observation error: the kernel-boundary, overshoot, and sub-threshold terms of slide 11, plus timestamp estimation and binning. The stability bounds then translate a sensor-error bound into an estimate bound.

## Learned temporal regularization

A possible extension: learn $\mathbf{\Lambda}$ in an outer problem [Antil–Di–Khatri 2020]. For $\mathbf{\Lambda} \succ 0$ the lower-level map $\mathbf{\Lambda} \mapsto \mathbf{f}^*(\mathbf{\Lambda})$ is single-valued and smooth.

## Shared-scene and orbital sensing

Characterize the motion-invariant modes of $B_{\mathrm{SAR}}$ for a given trajectory (joint ESAR work [45]); apply to sparse orbital streams such as the ISS data.

**Takeaway.** Each direction starts from a quantity the dissertation controls: an observation error, a solution map, or a null space.

Dissertation: Chapter 5, Future Directions; physical connection (p. 24); Proposition 2.4.1. External paper cited above.

## From event changes to well-posed estimators

**Temporal:** control an unobserved baseline.

**Spatial:** combine directional constraints globally.

**Binocular:** separate inner uniqueness from correspondence.

The common mathematical question is:

**Which directions do the observations control, and which are selected by the model?**

Thank you. Discussion and additional proofs follow.

# Discussion guide

**Further detail:** time-surface regression and contrast maximization; the full stereo arc (Chapter 4); references.

**Observation and temporal inversion**

- Minimal uniqueness condition / reference level
- Measure-valued continuous-time limit
- Signal gain versus noise gain
- Kernels; metrics; real time; $\mathbf{\Lambda(y)}$ stability

**Flow analysis**

- Weak Euler equation and boundary condition
- Full proof, stability, and discrete sweeps
- Exact block-Jacobi spectral argument
- Anisotropy; $\lambda, \alpha \to 0$; event-free regions
- Regression, projection, evaluation; splines
- Straight-edge aperture model

**Stereo analysis**

- Chapter 4 results summary
- Mean-zero conditioning
- Cost ellipsoid and observation perturbations
- IRLS contraction proof
- Fixed points versus likelihood stationarity
- Implemented scoring and aggregation

**Evidence and interpretation**

- Depth coverage; alignment; sensitivity
- One lemma behind the uniqueness results
- Contribution; parameters; learning
- Literature anchors and source map

# Must every temporal penalty entry be positive?

For arbitrary $\boldsymbol{\Lambda}$, the exact algebraic criterion is

$$\mathbf{A}^\top \mathbf{A} + \boldsymbol{\Lambda}^\top \boldsymbol{\Lambda} \succ 0 \quad \Longleftrightarrow \quad \ker \mathbf{A} \cap \ker \boldsymbol{\Lambda} = \{0\}.$$

## Why?

The quadratic form vanishes precisely when both $\mathbf{A}\mathbf{z} = 0$ and $\boldsymbol{\Lambda}\mathbf{z} = 0$.

On a connected first-difference chain, $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$.

- A single nonzero diagonal penalty entry removes that constant mode.
- Fixing one reference value or imposing zero mean also removes it.

**Takeaway.** Any choice of reference level resolves the mathematical ambiguity; a measured reference would add physical information.

Dissertation: Lemma 2.2.4; discussion following Lemma 4.4.1.

# The event measure has a jump primitive

At one pixel, set $\mu = \sum_j p_j \delta_{s_j}$, and choose a consistent right-continuous primitive

$$g(t) = f_0 + \mu((t_0, t]), \qquad g(b) - g(a) = \mu((a, b]).$$

For a standard smooth compactly supported unit-mass kernel,

$$e_h = \mu * \phi_h, \qquad g_h(t) = f_0 + \int_{t_0}^{t} e_h(\xi)\, d\xi.$$

For finitely many interior events, $g_h \to g$ in $L^1$ as $h \downarrow 0$.

- $g$ belongs to $BV$; a nonzero interior jump prevents $g \in H^1$.
- The displayed $g$ is an unregularized cumulative-event surrogate.

**Takeaway.** Keep interval conventions consistent and use increments, rather than an unscaled derivative, in $\mathbf{Af} \approx \mathbf{y}$.

Dissertation: Continuous-time formulation (p. 21); endpoint convention made explicit here.

# Latent-signal gain and observation-noise gain are different

For the unweighted monocular model with scalar $\lambda > 0$,

$$\mathbf{f}^* = (\mathbf{A}^\top \mathbf{A} + \lambda^2 \mathbf{I})^{-1} \mathbf{A}^\top \mathbf{y}.$$

| Input | Output modal gain | Interpretation |
|---|---|---|
| Latent signal | $g_k/(g_k + \lambda^2)$ | Low modes attenuated most |
| Difference observation | $\sqrt{g_k}/(g_k + \lambda^2)$ | Noise amplitude transfer |
| White difference noise | $\sigma^2 g_k/(g_k + \lambda^2)^2$ | Reconstructed modal variance |

$$\max_{g \geq 0} \frac{g}{(g + \lambda^2)^2} = \frac{1}{4\lambda^2}.$$

Dissertation: Theorem 4.6.21, Proposition 4.6.23.

# Kernel performance depends on the scene and adaptive weights

| | MVSEC indoor | | MVSEC outdoor | |
|---|---|---|---|---|
| Kernel | PSNR | CN | PSNR | CN |
| B-spline | 8.86 | 22.16 | 6.72 | 97.92 |
| Delta | 7.70 | 1375.75 | 7.97 | 53.34 |
| Gaussian | 8.84 | 20.22 | 6.81 | 98.90 |
| Rectangular | 8.24 | 423.59 | 7.20 | 148.33 |

- The temporal operator stays the same; represented data and adaptive penalties change.
- Better conditioning is not the same as smaller reconstruction bias.
- Reference-image scores do not identify an absolute intensity baseline from events.

Dissertation: Table 2.4, Table 2.5.

# What the reconstruction scores measure

| Quantity | What it compares | What it establishes |
|---|---|---|
| PSNR / SSIM | Reconstruction and reference images | Image agreement under the scoring convention |
| TC / TG | Temporal changes in the output | Smoothness of the returned sequence |
| ERC | Magnitudes of reconstructed changes and represented events | Correlation with event activity |
| CN | Sensitivity of a specified matrix | A numerical conditioning diagnostic |

## A simple check on interpretation

A constant sequence has $\mathrm{TC} = 1$ and $\mathrm{ERC} = 0$ under the stated conventions.

**Takeaway.** Consider reference fidelity, event consistency, and conditioning together. Smoothness alone rewards a static output.

Dissertation: Evaluation metrics (p. 31); Table 2.3; SSIM [40].

# Can the reconstruction run in real time?

**Practical experience:** a native C++/CUDA implementation used for live reconstruction in iniVation DV.

**Why live processing is feasible**

- Pixel histories give independent temporal problems.
- First differences give sparse temporal systems.
- GPU computation supports repeated reconstruction of incoming event batches.

**The processing budget**

- Active pixels and event workload.
- Number of temporal samples per reconstruction.
- Event-window length and output cadence.

These choices balance temporal detail, event context and responsiveness.

**Takeaway.** The live implementation supports near-real-time use; workload and temporal settings determine the operating point.

Source: Native DV module, C++ dispatcher, CUDA engine and build target (reconstruction_live_demo).

# The weak equation supplies the natural boundary condition

For every $\boldsymbol{\eta} \in H^1(\Omega;\mathbb{R}^2)$,

$$\int_\Omega (\mathbf{B}\mathbf{v}^* - b)\cdot\boldsymbol{\eta} + \alpha^2 \int_\Omega \nabla\mathbf{v}^* : \nabla\boldsymbol{\eta} = 0.$$

Interior tests give the distributional equation

$$-\alpha^2\Delta\mathbf{v}^* + \mathbf{B}\mathbf{v}^* = b.$$

Since the coefficients imply $\Delta\mathbf{v}^* \in L^2$, Green's identity gives

$$\partial_n\mathbf{v}^* = 0 \quad \text{in } H^{-1/2}(\partial\Omega;\mathbb{R}^2).$$

**Takeaway.** The theorem uses $H^1$ fields and weak traces, without claiming a classical solution on a Lipschitz domain.

Dissertation: Theorem 3.3.9, Theorem 3.4.2, and Theorem 3.4.3.

# Proof: control the mean and the fluctuation separately

Decompose $\mathbf{v} = \underbrace{\bar{\mathbf{v}}}_{\text{common drift (pan)}} + \underbrace{\mathbf{w}}_{\text{variation across the image}}$, with $\int_\Omega \mathbf{w} = 0$.

$$\|\mathbf{w}\|_{L^2} \le C_P \, \|\nabla \mathbf{v}\|_{L^2} .$$

Let $\beta_B = \|\mathbf{B}\|_{L^\infty}$ and $m = \lambda_{\min}(\int_\Omega \mathbf{B}) > 0$ (how well the edges cover every direction).

$$m|\bar{\mathbf{v}}|^2 \le 2 \int_\Omega \mathbf{v}^\top \mathbf{B} \mathbf{v} + 2\beta_B C_P^2 \, \|\nabla \mathbf{v}\|_{L^2}^2 .$$

Therefore the bilinear form satisfies

$$\mathcal{B}(\mathbf{v}, \mathbf{v}) = \int \mathbf{v}^\top \mathbf{B} \mathbf{v} + \alpha^2 \int |\nabla \mathbf{v}|^2 \ge c_0 \, \|\mathbf{v}\|_{H^1}^2 , \qquad c_0 = c_0(\Omega, \alpha, \beta_B, m).$$

**Takeaway.** Poincaré controls fluctuations, data control the mean, and Lax–Milgram supplies the unique weak solution.

Dissertation: proof of Theorem 3.6.1; Lax–Milgram [72].

# Stability follows by subtracting weak equations

Suppose two coefficient pairs share bounds

$$\|\mathbf{B}_i\|_{L^\infty} \le \beta_B, \qquad \lambda_{\min}\left(\int \mathbf{B}_i\right) \ge m > 0, \qquad \|b_i\|_{L^2} \le R_b.$$

Here $c_0 = c_0(\Omega, \alpha, \beta_B, m)$ is independent of $b_i$. For $\mathbf{z} = \mathbf{v}_1^* - \mathbf{v}_2^*$,

$$\mathcal{B}_1(\mathbf{z}, \mathbf{z}) = \int (b_1 - b_2) \cdot \mathbf{z} + \int \mathbf{z}^\top (\mathbf{B}_2 - \mathbf{B}_1)\mathbf{v}_2^*, \quad \text{hence}$$

$$\|\mathbf{z}\|_{H^1} \le c_0^{-1} \|b_1 - b_2\|_{L^2} + c_0^{-2} R_b \|\mathbf{B}_1 - \mathbf{B}_2\|_{L^\infty}.$$

In scene terms: jitter, binning, and noise events change the coefficients; constants grow as $m$ shrinks (stripes, slide 28). **Accuracy:** the bound concerns the minimizer of the chosen energy; its distance from the true motion depends on coefficient accuracy and on $H^1$ smoothing across motion boundaries (known-motion AEE, page 76).

**Takeaway.** The estimate is Lipschitz in its coefficients under uniform identifiability.

Dissertation: Theorem 3.6.1, Eq. 3.61.

# Discrete flow: connectivity and sampled directions matter

Fix $\alpha > 0$, $q_i > 0$, and $\mathbf{B}_i = \mathbf{B}_i^\top \succeq 0$ on a connected graph with $w_{ij} = w_{ji} \geq 0$ and degree $d_i = \sum_j w_{ij} > 0$.

## One simultaneous block-Jacobi sweep

$$\underbrace{(q_i \mathbf{B}_i + \alpha^2 d_i \mathbf{I}_2)}_{\text{local } 2\times 2 \text{ matrix}} \mathbf{v}_i^{(k+1)} = q_i b_i + \alpha^2 \sum_j w_{ij} \mathbf{v}_j^{(k)}.$$

- At each node, solve for two velocity components using local data and the previous neighbor velocities.
- Update every node from the same old field; then repeat the sweep.
- Scene terms: each pixel blends its edge measurement with neighbors' motion.

If $\sum_i q_i \mathbf{B}_i \succ 0$, the discrete energy has a unique minimizer, and these sweeps converge to it from every initial field.

Dissertation: Theorem 3.7.1 and Corollary 3.7.2; extension to positive node factors $q_i$.

# Why block-Jacobi converges for this graph model

Absorb $q_i$ into $\widehat{\mathbf{B}}_i = q_i \mathbf{B}_i \succeq 0$. Set

$$D_i = \alpha^2 d_i \mathbf{I} + \widehat{\mathbf{B}}_i, \qquad R_{ij} = \alpha^2 w_{ij} \mathbf{I}.$$

For symmetric nonnegative edge weights,

$$\mathbf{z}^\top (D \mp R)\mathbf{z} = \sum_i \mathbf{z}_i^\top \widehat{\mathbf{B}}_i \mathbf{z}_i + \alpha^2 \sum_{\{i,j\}} w_{ij} \left\| \mathbf{z}_i \mp \mathbf{z}_j \right\|^2 .$$

Connectedness and $\sum_i \widehat{\mathbf{B}}_i \succ 0$ eliminate both constant and alternating constant zero modes (the latter on bipartite graphs).

$$D - R \succ 0, \quad D + R \succ 0 \quad \Rightarrow \quad -\mathbf{I} \prec D^{-1/2} R D^{-1/2} \prec \mathbf{I}.$$

**Takeaway.** The Jacobi iteration matrix is similar to a symmetric matrix with spectral radius strictly below one.

Dissertation: extension of Theorem 3.7.1 and Corollary 3.7.2 by absorbing $q_i$ into $\widehat{\mathbf{B}}_i$. $\alpha > 0$, $d_i > 0$.

# Anisotropy needs a uniform ellipticity floor

For the normalized Nagel–Enkelmann tensor [67], write $g = \nabla \rho_{\mathrm{evt}}$ and $\gamma_{\mathrm{NE}} > 0$:

$$\mathbf{D} = \frac{g^{\perp}(g^{\perp})^{\top} + \gamma_{\mathrm{NE}}^2 \mathbf{I}}{|g|^2 + \gamma_{\mathrm{NE}}^2}.$$

Its tangent and normal eigenvalues are

$$1, \qquad \frac{\gamma_{\mathrm{NE}}^2}{|g|^2 + \gamma_{\mathrm{NE}}^2}.$$

If $g \in L^{\infty}$,

$$\theta \mathbf{I} \preceq \mathbf{D} \preceq \mathbf{I}, \qquad \theta = \frac{\gamma_{\mathrm{NE}}^2}{\|g\|_{L^{\infty}}^2 + \gamma_{\mathrm{NE}}^2} > 0.$$

**Takeaway.** Diffusion across event-density edges is reduced, while coercivity survives with $\alpha^2\theta$.

Dissertation: normalized variant of Eq. 3.46, Lemma 3.4.6, Corollary 3.6.4.

# A variable diffusion tensor must stay inside the divergence

For differentiable $\mathbf{D}$,

$$\begin{aligned}\nabla\cdot(\mathbf{D}\nabla u) = {} & D_{11}u_{xx} + 2D_{12}u_{xy} + D_{22}u_{yy} \\ & + (\partial_x D_{11} + \partial_y D_{12})u_x + (\partial_x D_{12} + \partial_y D_{22})u_y.\end{aligned}$$

Take $\rho_{\mathrm{evt}} = x^2/2$, $\gamma_{\mathrm{NE}} = 1$, and $u = x$:

$$\mathbf{D} = \operatorname{diag}\left(\frac{1}{1+x^2}, 1\right), \qquad \nabla\cdot(\mathbf{D}\nabla u) = \frac{-2x}{(1+x^2)^2} \not\equiv 0.$$

All second derivatives of $u$ vanish, so the principal part alone gives the wrong answer.

Dissertation: Anisotropic numerical discretization (p. 83).

# The regression radius formula is a heuristic

For full disks on a square lattice of spacing $h$, with $r/h$ large,

$$Q \sim \frac{\pi r^4}{4h^2}\mathbf{I}, \qquad U_h(r) = \frac{4}{81}L_H^2 r^4 + \frac{8}{\pi}\sigma^2 h^2 r^{-4}.$$

The minimum of this leading expression, when all parameters are positive, is

$$r^* = (162/\pi)^{1/8}(\sigma h/L_H)^{1/4}.$$

- The full disk must lie in a region with the asserted smoothness bound.
- $U_h$ is an asymptotic expression for the finite-design bound, not itself a proved bound on every discrete MSE.
- Affine time surfaces have zero curvature bias; this model supplies no finite balancing radius there.

Dissertation: Proposition 3.3.7 and Radius selection as a heuristic.

# Uniform projection and OLS slope need not be transport partners

The continuous OLS temporal slope over $[s, s+T]$ is

$$\widehat{\beta} = \int_s^{s+T} w_s(t) V_t(t)\, dt, \qquad w_s(t) = \frac{6(t-s)(s+T-t)}{T^3}.$$

Its transport-consistent spatial partner is the same $w_s$-weighted projection. Pairing it with a uniform projection gives residual

$$\mathbf{v} \cdot \nabla \int_s^{s+T} (T^{-1} - w_s(t)) V(\mathbf{x}, t)\, dt.$$

For $V(x,t) = (2 + x - t)^3$, $\mathbf{v} = (1, 0)$, $s = 0$, $T = 1$, this residual is $1/10$.

**Takeaway.** Well-posedness of a fixed-coefficient energy does not certify derivative consistency.

Dissertation: Section 3.4.1, consistent derivative-operator discussion.

# Mean-zero conditioning removes the unobserved mode

In the unweighted scalar model, restrict to

$$\mathcal{V}_0 = \{(\mathbf{f}_L, \mathbf{f}_R) : \mathbf{1}^\top \mathbf{f}_L = \mathbf{1}^\top \mathbf{f}_R = 0\}.$$

The sum–difference transform preserves this space. Its remaining eigenvalues are

$$g_k + \lambda^2, \qquad g_k + \lambda^2 + 2\gamma, \qquad k = 1, \dots, n-1.$$

Thus

$$\kappa_2(\mathbf{M}|_{\mathcal{V}_0}) = \frac{g_{\max} + \lambda^2 + 2\gamma}{g_1 + \lambda^2} \le \frac{g_{\max} + 2\gamma}{g_1}, \qquad g_1 = 4\sin^2 \frac{\pi}{2n}.$$

**Takeaway.** The restriction is SPD even at $\lambda = 0$. The uniform-in-$\lambda$ bound depends on the temporal grid; full-space perturbations remain different.

Dissertation: Corollary 4.6.17; Proposition 4.6.18 (notes).

# The selection theorem has an exact ellipsoid interpretation

Let

$$\mathbf{S} = \tfrac{1}{2}(\mathbf{I} - \mathbf{A}\mathbf{K}^{-1}\mathbf{A}^\top), \quad \mathbf{T} = \tfrac{1}{4}\mathbf{A}[\mathbf{K}^{-1} - (\mathbf{K} + 2\gamma\mathbf{I})^{-1}]\mathbf{A}^\top, \quad \mathbf{P} = \mathbf{S} + \mathbf{T} \succ 0.$$

Assume the ideal match $\mathbf{y}_{d_*} = \mathbf{y}_L$. For $\mathbf{e}_d = \mathbf{y}_L - \mathbf{y}_d$,

$$C(d) - C(d_*) = \|\mathbf{e}_d\|_{\mathbf{P}}^2 - 2\mathbf{y}_L^\top \mathbf{S}\mathbf{e}_d = \|\mathbf{y}_d - \mathbf{c}\|_{\mathbf{P}}^2 - \rho^2,$$

$$\mathbf{c} = \mathbf{P}^{-1}\mathbf{T}\mathbf{y}_L, \qquad \rho^2 = \mathbf{y}_L^\top \mathbf{S}\mathbf{P}^{-1}\mathbf{S}\mathbf{y}_L.$$

**Takeaway.** The ideal match wins if every competitor is outside the ellipsoid. In particular, $\|\mathbf{e}_d\|_{\mathbf{P}} > 2\rho$ is sufficient.

Dissertation: Proposition 4.6.8(a), (b2). Candidate-local unweighted model.

# An observation perturbation bound can protect a disparity winner

Keep $\mathbf{y}_L$, the candidate set, $\mathbf{A}$, $\mathbf{\Lambda}$, and $\gamma$ fixed. Perturb each right observation by $\boldsymbol{\xi}_d$. The quadratic candidate cost has

$$\nabla_{\mathbf{y}_d} C = 2\mathbf{P}(\mathbf{y}_d - \mathbf{c}), \qquad \nabla^2_{\mathbf{y}_d} C = 2\mathbf{P}, \qquad \|\mathbf{P}\| \leq \frac{1}{2}.$$

Consequently a uniform bound is

$$\eta = \max_d \left( \|\mathbf{y}_d - \mathbf{c}\| \, \|\boldsymbol{\xi}_d\| + \frac{1}{2} \|\boldsymbol{\xi}_d\|^2 \right).$$

If $\eta < \Delta/2$, the unique winning disparity is preserved.

**Takeaway.** The ideal correct match may be perturbed too. Changing the left data or adaptive parameters needs additional terms.

Dissertation: Corollary 4.6.10.

# How the reciprocal IRLS contraction bound is obtained

Write $\mathbf{B}_{\text{time}} = \operatorname{diag}(\mathbf{A}, \mathbf{A})$, $\mathbf{f} = \Phi(\mathbf{g})$, $\mathbf{f}' = \Phi(\mathbf{g}')$, and $\mathbf{r}' = \mathbf{y} - \mathbf{B}_{\text{time}}\mathbf{f}'$. Set $\mathbf{s} = \mathbf{B}_{\text{time}}\mathbf{g}$ and $\mathbf{s}' = \mathbf{B}_{\text{time}}\mathbf{g}'$. Subtract the frozen systems:

$$\mathbf{f} - \mathbf{f}' = \mathbf{M}_W^{-1}\mathbf{B}_{\text{time}}^{\top}\mathbf{W}^{1/2}\left[\mathbf{W}^{-1/2}(\mathbf{W} - \mathbf{W}')\mathbf{r}'\right].$$

Use

$$\left\|\mathbf{M}_W^{-1}\mathbf{B}_{\text{time}}^{\top}\mathbf{W}^{1/2}\right\| \le \ell^{-1/2}, \qquad \left\|\mathbf{W}'^{1/2}\mathbf{r}'\right\| \le \left\|\mathbf{W}'^{1/2}\mathbf{y}\right\|.$$

For reciprocal weights, $|w_i - w_i'| \le |s_i - s_i'|w_i w_i'$ and $w_i, w_i' \le \varepsilon^{-1}$.

**Takeaway.** Combining operator, residual, and weight-increment bounds gives $q = \|\mathbf{A}\|_2 \|\mathbf{y}\| / (\sqrt{\ell}\,\varepsilon^{3/2})$.

Dissertation: proof of Proposition 4.6.13; unclipped reciprocal map.

# A converged frozen-weight iteration need not be maximum likelihood

Consider the scalar Gaussian proxy with $y > 0$ and $\varepsilon > 0$, **without a regularizer**:

$$v(\theta) = |\theta| + \varepsilon, \qquad Q_k(\theta) = \frac{(y-\theta)^2}{2v(\theta_k)}.$$

Every frozen quadratic is minimized at $\theta = y$: the IRLS map has that fixed point. The state-dependent negative log-likelihood is

$$\mathcal{L}(\theta) = \frac{(y-\theta)^2}{2v(\theta)} + \frac{1}{2}\log v(\theta).$$

Yet $\mathcal{L}'(y) = 1/[2(y+\varepsilon)] > 0$.

**Takeaway.** The frozen-weight fixed point is not even stationary for this proxy likelihood.

Dissertation: Proposition 4.6.2. Scalar logical illustration outside the positive-$\Lambda$ standing assumptions.

# What the implemented IRLS procedure returns

At every candidate disparity:

1. Initialize from the unweighted exact solve.
2. Recompute predicted-rate weights, **clip for the solve**, and freeze them.
3. Solve the SPD system and damp toward its minimizer.
4. Recompute **unclipped scoring weights**; apply acceptance and stopping rules.
5. Return the last accepted iterate and its score.

The score omits the log-variance term. Masking, averaging, margin normalization, and SGM [95] then change candidate comparisons.

**Takeaway.** Brouwer and the contraction theorem describe fixed maps; early stopping and aggregation require their own analysis.

Dissertation: Section 4.5, Section 4.6.1; Section A.2.

# Depth evaluation distinguishes returned coverage and scored coverage

1. Ground truth is finite and in the reporting depth band.
2. The returned estimate is finite.
3. The estimate also lies in the reporting band.

| Configuration | Returned coverage | In-band retention | Scored points |
|---|---|---|---|
| Control | 50.17% | 51.54% | 1.43 M |
| WLS | 49.48% | 96.40% | 2.66 M |

The scoring band is $[0.5, 8]$ m. Out-of-band finite depths are excluded by convention; they are not automatically physically invalid.

Dissertation: Table A.4, Weighting comparison (p. 182). Complete configurations; method-specific populations.

# Pooled kurtosis does not establish heavy-tailed active errors

Suppose a residual distribution mixes a point mass at zero with a zero-mean Gaussian of variance $\sigma^2 > 0$, with zero-mass probability $0 \leq \pi_0 < 1$.

$$\mathbb{E}X^2 = (1-\pi_0)\sigma^2, \qquad \mathbb{E}X^4 = 3(1-\pi_0)\sigma^4.$$

Its population excess kurtosis is

$$\frac{\mathbb{E}X^4}{(\mathbb{E}X^2)^2} - 3 = \frac{3}{1-\pi_0} - 3.$$

- Event-free pixels can have exactly zero fitted transport residual.
- Regularization, unequal variances, and temporal dependence affect diagnostics.
- Test event-active and variance-standardized fitted residuals before choosing a tail model.

Dissertation: Section 3.8.1; Model Mismatch in the Residuals.

# Adaptive temporal weights are fixed in the theorem

For event activity $a_i = |dc_i|$, Chapter 2 uses these positive rules:

$$\lambda_i = 10^{-3} + \frac{10}{1 + e^{a_i}} \qquad \text{(sigmoid)}$$

$$\lambda_i = \max\left(10^{-3}, \min\left(\frac{100}{10^{-3} + a_i}, 100\right)\right) \qquad \text{(inverse proportional)}$$

$$\lambda_i = 10^{-3} + 10e^{-\min(a_i, 10)} \qquad \text{(exponential decay)}$$

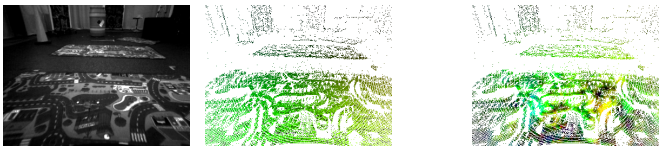
Chapter 4 instead uses $\lambda_{xy} = \max\{\lambda_{\text{floor}}, \lambda_0 \|\mathbf{b}\| / (\|\mathbf{b}\| + 2)\}$.

**Takeaway.** Each fixed $\mathbf{\Lambda}$ is covered by the theorem; for the Chapter 2 rules, the data-dependent map $\mathbf{y} \mapsto \mathbf{\Lambda}(\mathbf{y}) \mapsto \mathbf{f}^*$ is locally Lipschitz (p. 98).

Dissertation: Adaptive temporal weights (p. 38); Section 4.5 (regularization terms); Theorem 2.2.5.

# A real MVSEC flow example is qualitative

MVSEC indoor_flying2, frame 0340.




**Takeaway.** This is one qualitative example; Chapter 3 reports no aggregate real-data AEE.

Dissertation: Figure 3.2.

# Quadratic $H^1$ regularization and TV answer different questions

The flow energy uses a quadratic gradient penalty:

$$E(\mathbf{v}) = \tfrac{1}{2}\int_\Omega \left(\mathbf{v}^\top \mathbf{B}\mathbf{v} - 2b^\top \mathbf{v}\right) + \tfrac{\alpha^2}{2}\,\|\nabla \mathbf{v}\|_{L^2}^2\,, \qquad \mathbf{v} \in H^1(\Omega;\mathbb{R}^2).$$

- Lax–Milgram gives existence, uniqueness, and coefficient stability under the stated hypotheses.
- TV [41] can preserve jumps, but needs a separate $BV$ existence argument and its own uniqueness and stability assumptions.
- Normalized NE diffusion gives directional quadratic smoothing with a uniform ellipticity floor.

**Takeaway.** The integrated full-rank condition supports the quadratic $H^1$ proof; it does not establish a TV theorem.

Dissertation: Theorem 3.6.1, function-space remark, Eq. 3.46.

# Frozen prediction weights are not generally an MM majorizer

For a scalar prediction $s$, let $v(s) = |s| + \varepsilon$, $\varepsilon > 0$, and

$$S(s) = \frac{(y-s)^2}{2v(s)} + R(s), \qquad Q_k(s) = \frac{(y-s)^2}{2v(s_k)} + R(s).$$

The frozen surrogate matches $S$ at $s_k$, but away from $s = 0$ its derivative omits

$$-\frac{v'(s)(y-s)^2}{2v(s)^2}.$$

**Takeaway.** Without a tangent derivative, the frozen quadratic is not generally a differentiable tangent majorizer.

Dissertation: Section 4.6.1, Proposition 4.6.2; derivative consequence, not a stated MM theorem.

# A Gaussian model gives an evidence interpretation of attenuation

For common fixed $\mathbf{\Lambda} \succ 0$, $\gamma \geq 0$, and independent unit Gaussian noise, the candidate conditional is

$$\mathbf{y}_R \mid \mathbf{y}_L \sim \mathcal{N}(\mathbf{c}, (2\mathbf{P})^{-1}), \qquad \mathbf{c} = \mathbf{P}^{-1}\mathbf{T}\mathbf{y}_L.$$

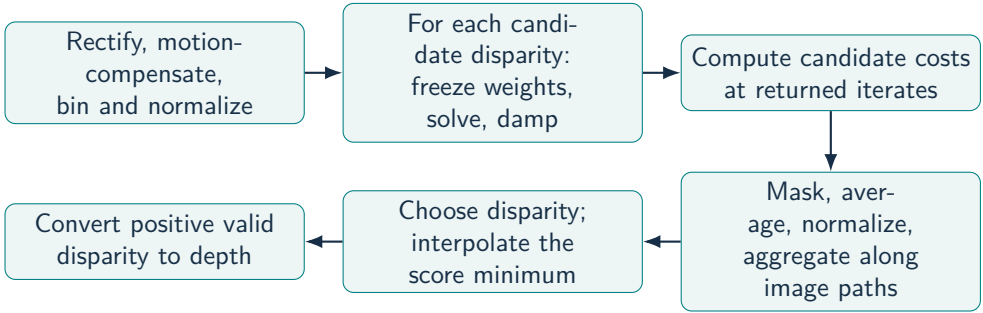
Coupled versus independent evidence satisfies

$$-\log \frac{p_\gamma(\mathbf{y}_L, \mathbf{y}_R)}{p_0(\mathbf{y}_L, \mathbf{y}_R)} = D(\mathbf{y}_L - \mathbf{y}_R) + \text{constant}.$$

**Takeaway.** The conditional mean explains attenuation; the discrepancy alone is an unevaluated alternative score.

Dissertation: Supplementary Gaussian derivation from Lemma 4.6.6, Proposition 4.6.8.

# The implemented estimator has several distinct layers




Theorems describe specific layers; empirical evaluation describes the complete estimator.

**Takeaway.** The exact minimum of a computed score list is not a certificate of the true disparity or of a global image-level energy minimum.

Dissertation: Section 4.5; Section A.2.

# What does the dissertation contribute mathematically?

- A transparent temporal inversion with an explicit observation null space and quantitative fixed-weight guarantees.
- One well-posedness and stability theorem for several event-derived quadratic flow energies, with representation-specific hypotheses.
- A distinction between a coercive spline regularizer and a nonconvex contrast objective, plus full-objective existence.
- Stereo block structure, profiled-cost bias, ideal selection and robustness criteria, and a scoped adaptive-weight analysis.

**Takeaway.** Classical proof tools connect the models; the event-specific formulation, hypotheses, consequences, and limitations are the research content.

Dissertation: Section 5.3 and Table 5.1.

# Literature anchors and where to find the claims

**Image reconstruction**

- Rebecq et al. [12]
  *Events-to-Video.*
- Zhang, Yezzi & Gallego [6]
  *Image reconstruction as a linear inverse problem using optical flow.*
- Antil–Sayre [2]
  *Bilevel inverse problems in neuromorphic imaging.*

**Sensor overview**

Gallego et al. [1]: *Event-Based Vision: A Survey.*

**Flow models and analysis**

- Horn–Schunck [16]
  *Determining Optical Flow.*
- Benosman et al. [50]
  *Event-Based Visual Flow.*
- Weickert–Schnörr [57]
  *Convex regularizers for image motion.*
- Gallego, Rebecq & Scaramuzza [3]
  *A Unifying Contrast Maximization Framework.*

**Evaluation data**

Zhu et al. [35]: *The Multivehicle Stereo Event Camera Dataset.*

Dissertation: Chapter 2; Section 3.2; Section A.2. Numbers match the dissertation bibliography; full entries on the closing reference pages.

# Stable estimates can still have regularization bias

For fixed scalar $\lambda > 0$, $\mathbf{y} = \mathbf{A}\mathbf{f}^\dagger + \boldsymbol{\eta}$, and $\mathbf{K} = \mathbf{A}^\top\mathbf{A} + \lambda^2\mathbf{I}$,

$$\mathbf{f}_\lambda - \mathbf{f}^\dagger = -\lambda^2\mathbf{K}^{-1}\mathbf{f}^\dagger + \mathbf{K}^{-1}\mathbf{A}^\top\boldsymbol{\eta}.$$

The two terms are regularization bias and propagated observation noise.

**Takeaway.** A stable estimate need not equal the physical signal.

Dissertation: Theorem 2.2.5, Proposition 2.2.7; direct bias-noise identity.

# The physical event stream loses more than one baseline

The binned increment matrix has $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$, but the ideal sensor can lose time-varying signals too.

$$L_1(t) = L_0, \qquad L_2(t) = L_0 + a\sin(2\pi t/T), \qquad 0 < a < c_{\mathrm{th}}, \quad t \in [0, T].$$

Both start at $L_0$, emit no events, and have positive intensities $I_j(t) = e^{L_j(t)}$.

**Takeaway.** The chain nullspace describes the discrete increment surrogate, not every ambiguity of the physical sensor.

Dissertation: Eq. 2.1, Lemma 2.2.4; direct ideal-sensor example.

# Integrated directional information is necessary for uniqueness

Let $\Omega$ be connected, bounded, and Lipschitz; $\alpha > 0$; $\mathbf{B} \in L^\infty$ symmetric PSD; and $b \in L^2$. Minimize over unrestricted $H^1(\Omega; \mathbb{R}^2)$ fields. Set $M_B = \int_\Omega \mathbf{B}$. If $M_B$ is singular, choose $c \neq 0$ with $M_B c = 0$. Then $\mathbf{B}c = 0$ a.e. and

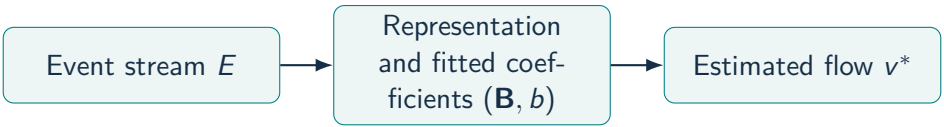
$$E(\mathbf{v} + tc) = E(\mathbf{v}) - t\, c^\top \int_\Omega b.$$

If the coefficient is nonzero, $E$ is unbounded below; if zero, it is flat in direction $c$.

**Takeaway.** The integrated matrix condition is necessary for a unique minimizer in this unrestricted quadratic model.

Dissertation: Theorem 3.6.1; direct converse for its quadratic energy.

# Coefficient stability is one stage of event-to-flow stability




The theorem controls coefficient-to-flow sensitivity. A Lipschitz representation map could compose with that bound.

**Takeaway.** An end-to-end guarantee also needs a metric and stability result for event preprocessing.

Dissertation: Theorem 3.6.1, Proposition 3.3.7; supplementary scope explanation.

# Refinement requires a choice of continuum scaling

Illustrative smooth, uniform-time model only: spacing $h$, increments $y_i = hg_i$, and target energy

$$\frac{1}{2}\int |f' - g|^2 dt + \frac{\beta_{\text{time}}^2}{2}\int |f|^2 dt \quad \rightsquigarrow \quad \frac{1}{2h}\|\mathbf{A}f - y\|^2 + \frac{\beta_{\text{time}}^2 h}{2}\|f\|^2 .$$

Multiplying the discrete objective by $h$ gives the Chapter 2 form with

$$\lambda_h = \beta_{\text{time}} h.$$

**Takeaway.** The penalty's grid scaling depends on the continuum objective being approximated.

Dissertation: Eq. 2.7, Eq. 2.8; supplementary smooth-model scaling illustration.

# Strong smoothness selects one constant flow

For fixed $\Omega, \mathbf{B}, b$ satisfying the unified theorem, with unrestricted $H^1$ fields and the same $\alpha^2$ gradient coefficient,

$$\alpha \to \infty \quad \Longrightarrow \quad v_\alpha \to c \text{ in } H^1, \qquad c = \left( \int_\Omega \mathbf{B} \right)^{-1} \int_\Omega b.$$

**Takeaway.** Very strong spatial regularization removes variation and leaves the data-balanced constant field.

Dissertation: Theorem 3.6.1; direct strong-regularization limit.

# Transporting a scalar differs from conserving a density

Scalar transport and source-free density conservation are

$$V_t + v \cdot \nabla V = 0, \qquad V_t + \nabla \cdot (Vv) = 0, \qquad \nabla \cdot (Vv) = v \cdot \nabla V + V \nabla \cdot v.$$

They coincide when $V \nabla \cdot v = 0$, in particular for divergence-free flow.

**Takeaway.** The continuity equation adds a density-compression term to scalar transport.

Dissertation: Section 3.4.1, Scalar transport and density conservation (p. 85); direct product-rule comparison.

# Correlation changes uncertainty without removing coercivity

For a centered symmetric regression design $Z$ with $Q = Z^\top Z \succ 0$, timestamp-error covariance $\Sigma$ gives

$$\mathrm{Cov}(\hat{g}) = Q^{-1} Z^\top \Sigma Z Q^{-1}.$$

This reduces to $\sigma^2 Q^{-1}$ when $\Sigma = \sigma^2 I$.

With any $\Gamma \succ 0$, the temporal system matrix is $\mathbf{A}^\top \Gamma^{-1} \mathbf{A} + \mathbf{\Lambda}^2 \succ 0$ when $\mathbf{\Lambda} \succ 0$.

**Takeaway.** The deterministic quadratic solve stays coercive, while statistical uncertainty depends on the error covariance.

Dissertation: Proposition 3.3.7; Eq. 2.8, Theorem 2.2.5; direct correlated-error extension.

# Optical flow: further analysis

- Time-surface gradient estimation: bias, variance, and design.
- Contrast maximization: what it models, known-motion evidence, objective, and existence proof.

Regression quantifies coefficient uncertainty; contrast maximization changes the objective being optimized.

# Time-surface gradient estimation: bias versus variance

Fit a plane to timestamps at offsets $\mathbf{z}_j$ around a pixel.

$$Q = \sum_j \mathbf{z}_j \mathbf{z}_j^\top \succ 0.$$

For a centrally symmetric design and independent timestamp errors with common mean and finite variance $\sigma^2$:

$$\mathrm{Cov}(\widehat{\nabla T}) = \sigma^2 Q^{-1}.$$

With a Lipschitz Hessian of constant $L_H$ on a neighborhood containing the sample disk,

$$\|\mathrm{Bias}\| \le \frac{L_H}{6} \left\|Q^{-1}\right\| \sum_j \|\mathbf{z}_j\|^4, \qquad \mathrm{MSE} = \|\mathrm{Bias}\|^2 + \sigma^2 \,\mathrm{tr}(Q^{-1}).$$

**Takeaway.** A larger neighborhood reduces variance but can increase curvature bias. Regression rank and velocity identifiability are different questions.

Dissertation: Proposition 3.3.7.

# Contrast maximization finds the motion that sharpens the events

**What it models**

- A moving edge leaves a smeared trail of events over the time window.
- Slide each event back along a candidate flow to one reference time. The correct flow stacks the trail into a sharp edge; image variance measures the sharpness.
- It uses individual events in continuous time, with no time surface, bins, or fitted gradients. Binned models blur when an edge moves during a bin.

**Known-motion evidence**

AEE (px) at window 22; a zero flow field scores 3.00.

| Scene | Best other (struct. tensor) | CMAX |
|---|---|---|
| Idealized stripes | 2.15 | **0.49** |
| Simulated stripes (v2e) | 1.92 | **1.19** |
| Simulated checkerboard (v2e) | 0.98 | **0.32** |

On the stripes, CMAX recovers the diagonal motion: warping the events over the whole window brings in evidence from the object's boundary.

**Takeaway.** CMAX has the lowest error in every known-motion scene.

Dissertation: Section 3.5; Figure 3.3 and Figure 3.4; contrast maximization [3, 55]; v2e [46].

# How the known-motion comparison was evaluated

| Aspect | Reported protocol |
|---|---|
| Stimulus | A $60 \times 40$ px object translates diagonally at 3 px per window; direct idealized events and v2e-simulated events [46]. |
| Settings | Quadratic variants: $\alpha_{\mathrm{smooth}} = 20$, no pyramid. Contrast maximization: 100 L-BFGS [71] iterations. |
| Selection | Window 22 minimizes summed deviation from each variant's median AEE over windows 1–31. |
| Display / error | Common arrow scale; AEE is mean Euclidean displacement error, in pixels per evaluation window. |

Equal numerical $\alpha$ values give different stencil-dependent smoothing strengths.

**Takeaway.** These controlled examples show directional-information effects at the reported solver settings.

Dissertation: Figure 3.3 and Figure 3.4; smoothing stencils (p. 164).

# Contrast maximization changes the optimization problem

Warp each event to a common normalized reference time:

$$\mathbf{x}_k^w = \mathbf{x}_k + (\widetilde{t}_{\mathrm{ref}} - \widetilde{t}_k)\mathbf{q}(\mathbf{x}_k), \qquad \mathbf{q} = T\mathbf{v}_{\mathrm{phys}}.$$

Represent $\mathbf{q}$ by a tensor-product cubic B-spline on a uniform knot grid ($h = 8\,\mathrm{px}$), with coefficients $\mathbf{c}$; both components share the basis.

$$F(\mathbf{c}) = -\operatorname{Var}(I_{\mathbf{c}}) + \mathbf{c}^\top \mathbf{K}_{\mathrm{reg}}\mathbf{c}, \qquad \mathbf{K}_{\mathrm{reg}} \succeq \tau_{\mathrm{ridge}}\mathbf{I}, \quad \tau_{\mathrm{ridge}} > 0.$$

- A good warp concentrates aligned events.
- A flexible incorrect warp can also concentrate events by collapsing them [55].

**Takeaway.** The full objective is generally nonconvex. The quadratic flow theorem does not apply.

Dissertation: Section 3.5, Proposition 3.5.1, Proposition 3.5.2; contrast maximization [3].

# A global minimizer exists, even though the solver is local

For finitely many fixed event weights, let $W_{\mathrm{tot}} = \sum_k |w_k|$. The bilinear accumulation is continuous and satisfies

$$\mathrm{Var}(I_{\mathbf{c}}) \le \frac{W_{\mathrm{tot}}^2}{HW}.$$

Therefore

$$F(\mathbf{c}) \ge -\frac{W_{\mathrm{tot}}^2}{HW} + \tau_{\mathrm{ridge}} \|\mathbf{c}\|^2 \longrightarrow \infty.$$

## Proof mechanism

Continuous objective + bounded focus gain + coercive ridge $\Rightarrow$ compact sublevel set $\Rightarrow$ Weierstrass.

**Takeaway.** A global minimizer exists by finite-dimensional coercivity; the local solver has no global guarantee.

Dissertation: Proposition 3.5.2 and evaluated-solver scope.

# Stereo depth: full Chapter 4 arc

1. Coupled model and the frozen-weight inner solve.
2. Sum and difference modes; the profiled candidate cost.
3. The explicit counterexample and the correct-selection theorem.
4. Winning-margin stability and the non-identifiability example.
5. Heteroscedastic weights and IRLS fixed-point analysis.
6. MVSEC benchmark, weighting comparison, and hard fusion.

These results separate the inner temporal solve from the outer correspondence decision.

# Stereo depth in one slide: what Chapter 4 adds

At each candidate disparity $d$, solve two coupled temporal profiles:

$$\min_{\mathbf{f}_L,\mathbf{f}_R} \sum_{c=L,R} \left( \tfrac{1}{2} \left\| \mathbf{W}_c^{1/2} (\mathbf{A}\mathbf{f}_c - \mathbf{y}_c) \right\|^2 + \tfrac{1}{2} \left\| \mathbf{\Lambda} \mathbf{f}_c \right\|^2 \right) + \tfrac{\gamma}{2} \left\| \mathbf{f}_L - \mathbf{f}_R \right\|^2 .$$

1. **Inner solve.** The Hessian is $\succeq \lambda_{\min}^2 \mathbf{I}$: the temporal proof plus a nonnegative coupling term.
2. **Profiled cost.** $C(d) = m(\mathbf{y}_L) + m(\mathbf{y}_R(d)) + D_d$: the monocular term moves with $d$. In an exact two-sample example the perfect match costs $\frac{1}{3}$, an attenuated one $\frac{9}{40}$: a weaker right-view copy can outscore the true match.
3. **Selection.** A separation condition makes the correct candidate the unique winner; a margin $\Delta$ tolerates cost perturbations below $\Delta/2$ (e.g., noise events, threshold mismatch between sensors).
4. **Reweighting.** The IRLS map has a fixed point (Brouwer) and contracts under a sufficient condition.

MVSEC indoor flying: soft-coupled WLS is best on Flying 2 (58.35 cm) and log RMSE $\times 100$ (44.54), second elsewhere, with 2.66 M scored points versus 1.20 M for the strongest prior method.

**Takeaway.** Coupling keeps every inner solve well posed; the right correspondence also needs a cost margin.

Dissertation: Theorem 4.6.3, Proposition 4.6.7, Proposition 4.6.8, Proposition 4.6.13; Table A.5; MVSEC [35], Piatkowska [78], Brouwer [92].

# Stereo alignment after motion compensation

The model pairs synchronized, rectified observations of a scene point visible in both cameras.

## Each camera fits its own translation within each temporal slice

$$\mathbf{x}'_c = \mathbf{x}_c + (t_{\mathrm{ref}} - s)\mathbf{v}_c, \qquad c \in \{L, R\}.$$

For simultaneous rectified correspondences,

$$d_{\mathrm{w}}(s) = d(s) + (t_{\mathrm{ref}} - s)(v_{x,L} - v_{x,R}).$$

- A horizontal warp mismatch changes the disparity presented to the matcher.
- A vertical warp mismatch moves a correspondence away from its scanline.
- Reported slices are 12.5 ms wide within a 0.25 s evaluation window.

**Takeaway.** A unique inner solve fits the supplied observations; physical correspondence also relies on compatible timing, calibration, and warps.

Dissertation: Motion-compensated accumulation (p. 108); Table A.1.

# At each candidate, couple two temporal profiles

Suppress the pixel index; $\mathbf{y}_R = \mathbf{y}_R(d)$.

$$J_d(\mathbf{f}_L, \mathbf{f}_R; \mathbf{W}_L, \mathbf{W}_R) = \frac{1}{2}\left\|\mathbf{W}_L^{1/2}(\mathbf{A}\mathbf{f}_L - \mathbf{y}_L)\right\|^2 + \frac{1}{2}\left\|\mathbf{W}_R^{1/2}(\mathbf{A}\mathbf{f}_R - \mathbf{y}_R)\right\|^2 + \frac{1}{2}\left\|\mathbf{\Lambda}\mathbf{f}_L\right\|^2 + \frac{1}{2}\left\|\mathbf{\Lambda}\mathbf{f}_R\right\|^2 + \frac{\gamma}{2}\left\|\mathbf{f}_L - \mathbf{f}_R\right\|^2.$$

- Data fidelity estimates each temporal profile from its own view.
- Magnitude penalties control temporal baselines.
- Soft coupling penalizes left–right disagreement.

**Takeaway.** The right latent profile is auxiliary to this pixel and candidate, rather than a single globally shared right image.

Dissertation: Eq. 4.24; fixed-disparity model definition.

# Frozen weights give a unique, stable inner solve

Fix $d$, $\mathbf{W}_L, \mathbf{W}_R \succ 0$, $\mathbf{\Lambda} \succ 0$, and $\gamma \geq 0$. Define

$$\mathbf{K}_c = \mathbf{A}^\top \mathbf{W}_c \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda}, \qquad \mathbf{M}_W = \begin{pmatrix} \mathbf{K}_L + \gamma \mathbf{I} & -\gamma \mathbf{I} \\ -\gamma \mathbf{I} & \mathbf{K}_R + \gamma \mathbf{I} \end{pmatrix}.$$

For $\mathbf{z} = (\mathbf{z}_L, \mathbf{z}_R)$,

$$\mathbf{z}^\top \mathbf{M}_W \mathbf{z} = \sum_{c=L,R} \left( \left\| \mathbf{W}_c^{1/2} \mathbf{A} \mathbf{z}_c \right\|^2 + \left\| \mathbf{\Lambda} \mathbf{z}_c \right\|^2 \right) + \gamma \left\| \mathbf{z}_L - \mathbf{z}_R \right\|^2.$$

**Conclusion**

$\mathbf{M}_W \succeq \lambda_{\min}^2 \mathbf{I} \succ 0$: the frozen quadratic has exactly one global minimizer.

Dissertation: Eq. 4.34, Theorem 4.6.3, Theorem 4.6.11.

# Sum and difference modes expose the coupling

For equal blocks $\mathbf{K}_L = \mathbf{K}_R = \mathbf{K}$, introduce

$$\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{I} & -\mathbf{I} \end{pmatrix}, \qquad \mathbf{U}^\top \mathbf{M} \mathbf{U} = \begin{pmatrix} \mathbf{K} & 0 \\ 0 & \mathbf{K} + 2\gamma \mathbf{I} \end{pmatrix}.$$

**Mean profile**
The coupling cancels. Shared information is fused by the temporal solve.

**Difference profile**
The coupling raises curvature and suppresses disagreement.

For $\mathbf{\Lambda} = \lambda \mathbf{I}$ in the unweighted model,

$$\kappa_2(\mathbf{M}) = 1 + \frac{g_{\max} + 2\gamma}{\lambda^2}.$$

Dissertation: Lemma 4.6.15, Theorem 4.6.16.

# Profile out the latent variables to see what disparity compares

In the unweighted model, let $\mathbf{K} = \mathbf{A}^\top \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda}$ and

$$m(\mathbf{y}) = \frac{1}{2} \mathbf{y}^\top (\mathbf{I} - \mathbf{A}\mathbf{K}^{-1}\mathbf{A}^\top)\mathbf{y}.$$

The exact minimized candidate cost is

$$C(d) = m(\mathbf{y}_L) + m(\mathbf{y}_R(d)) + \frac{1}{4}\boldsymbol{\delta}_d^\top \left[\mathbf{K}^{-1} - (\mathbf{K} + 2\gamma\mathbf{I})^{-1}\right] \boldsymbol{\delta}_d, \quad \boldsymbol{\delta}_d = \mathbf{A}^\top(\mathbf{y}_L - \mathbf{y}_R(d)).$$

- The final term measures left–right discrepancy.
- The right monocular cost also changes with the candidate.

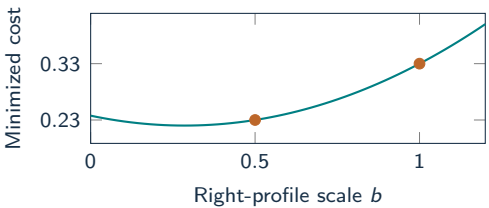
**Takeaway.** The minimized regularized energy is more than a correspondence discrepancy.

Dissertation: Lemma 4.6.6, Proposition 4.5.1. Equal fixed weights admit a scaled analogue.

# A perfect correspondence need not have the lowest cost

Take $\mathbf{A} = [-1, 1]$, $\mathbf{\Lambda} = \mathbf{I}$, $\gamma = 1$, $y_L = 1$, and $y_R = b$.

$$C(b) = \frac{1 + b^2}{6} + \frac{(1 - b)^2}{15}.$$




Let $D$ denote the discrepancy term in the preceding profiled-cost formula.

$$C(1) = \frac{1}{3}, \qquad C\left(\frac{1}{2}\right) = \frac{9}{40} < \frac{1}{3}.$$

$$b^* = \frac{D(y_L)}{m(y_L) + D(y_L)} = \frac{2}{7}.$$

An attenuated wrong candidate beats the exact match if both are available.

**Takeaway.** Inner uniqueness, distinct observations, and correct correspondence are separate claims.

Dissertation: Proposition 4.6.7, Correspondence identifiability (p. 143). Unweighted cost before aggregation.

# Correct disparity needs a separation condition

Write the unweighted cost as

$$C(d) = m(\mathbf{y}_L) + m(\mathbf{y}_d) + D(\mathbf{y}_L - \mathbf{y}_d), \qquad D(\mathbf{e}) \geq 0.$$

## A sufficient correct-selection theorem

On a finite candidate set, suppose $\mathbf{y}_{d_*} = \mathbf{y}_L$, the positive regularizer is common to both views and all candidates, and $\gamma > 0$ is candidate-independent. If every competitor has

$$\mathbf{y}_d \neq \mathbf{y}_L \quad \text{and} \quad m(\mathbf{y}_d) \geq m(\mathbf{y}_L),$$

then $C(d) > C(d_*)$: the correct candidate is the unique winner.

**Takeaway.** The discrepancy must overcome candidate-dependent monocular bias. Stronger metric-separation criteria are also available.

Dissertation: Proposition 4.6.8(b1); additional criterion in backup.

# A winning margin turns cost stability into selection stability

For a fixed finite set of at least two candidates with a unique winner $d_*$, define

$$\Delta = \min_{d \neq d_*} \left( C(d) - C(d_*) \right) > 0.$$

If $|\widetilde{C}(d) - C(d)| \leq \eta$ for all candidates, including $d_*$, then every $d \neq d_*$ satisfies

$$\widetilde{C}(d) - \widetilde{C}(d_*) \geq \Delta - 2\eta > 0 \qquad \text{when } \eta < \Delta/2.$$

## A non-identifiability example

Two candidates with identical right-view profiles have identical local costs when the candidate procedures and parameters coincide.

**Takeaway.** Temporal activity alone cannot guarantee correspondence separation.

Dissertation: Corollary 4.6.10, Proposition 4.6.20.

# Depth sensitivity grows with distance

For fixed calibration and positive $d$ and $d + \delta d$,

$$Z(d) = \frac{fB}{d}, \qquad |Z(d + \delta d) - Z(d)| = \frac{fB|\delta d|}{d(d + \delta d)}.$$

**Small relative disparity error**

$$|\delta Z| \approx \frac{Z^2}{fB}|\delta d|, \qquad \frac{|\delta Z|}{Z} \approx \frac{|\delta d|}{d}.$$

At the same disparity precision, absolute depth error grows quadratically with distance.

**Illustrative exact calculation**

| | Reference $d$ | Estimated $d$ |
|---|---|---|
| Farther point | 4 px | 5 px |
| Nearer point | 20 px | 21 px |

Depth is underestimated by 20% and 4.76%, respectively.

**Takeaway.** Read depth errors with the disparity precision, depth range, calibration, and scored population.

Dissertation: Eq. 4.11 and Eq. 4.12; stereo geometry [80]; example percentages are illustrative arithmetic.

# Heteroscedastic weights: statistical motivation and scope

As a working raw-count model,

$$N_+ \sim \text{Poisson}(\theta_+), \qquad N_- \sim \text{Poisson}(\theta_-),$$

independently. For $b = N_+ - N_-$ (Skellam [87]),

$$\mathbb{E}b = \theta_+ - \theta_-, \qquad \text{Var}(b) = \theta_+ + \theta_-.$$

$$|\theta_+ - \theta_-| \leq \theta_+ + \theta_-, \quad \text{with equality iff at least one rate is zero.}$$

The evaluated reciprocal proxy is

$$W_{ii}^{(k)} = \frac{1}{|(\mathbf{A}\mathbf{f}^{(k)})_i| + \varepsilon_{\text{noise}}}, \qquad \varepsilon_{\text{noise}} > 0.$$

**Takeaway.** Net rate does not determine total variance. The processed-data weights define a frozen Gaussian surrogate.

Dissertation: Definition 4.6.1 and following scope discussion.

# IRLS: fixed-point existence and conditional convergence

Fix $d$, $\mathbf{\Lambda}$, and $\gamma$; stack $\mathbf{y} = (\mathbf{y}_L, \mathbf{y}_R)$. Let $\Phi(\mathbf{g})$ minimize the quadratic with weights frozen at $\mathbf{g}$. For the *unclipped reciprocal map*, set $\ell = \lambda_{\min}^2 > 0$:

$$\|\Phi(\mathbf{g})\|^2 \leq \frac{\|\mathbf{y}\|^2}{\ell\, \varepsilon_{\text{noise}}}.$$

A sufficiently large closed ball is invariant; continuity and Brouwer [92] give a fixed point.

## Sufficient convergence condition

$$q = \frac{\|\mathbf{A}\|_2 \, \|\mathbf{y}\|}{\sqrt{\ell}\, \varepsilon_{\text{noise}}^{3/2}} < 1 \quad \Rightarrow \quad \text{unique fixed point and geometric convergence.}$$

For fixed damping $\omega \in (0, 1]$, the bound is $q_\omega = 1 - \omega + \omega q$.

**Takeaway.** A fixed point exists; convergence follows under the displayed sufficient condition.

Dissertation: Lemma 4.6.12, Proposition 4.6.13, Section 4.6.1; IRLS [84].

# Stereo benchmark: read accuracy alongside scored coverage

| Method | Mean depth error (cm) ↓ Fl1 | Fl2 | Fl3 | Fl4 | log RMSE ×100 ↓ | Scored points (M) |
|---|---|---|---|---|---|---|
| Unweighted control | 140.23 | 115.40 | 142.35 | 113.00 | 96.18 | 1.43 |
| Unweighted SGM [95] | 83.45 | 72.90 | 85.33 | 71.62 | 50.55 | 2.39 |
| **Soft-coupled WLS** | 73.83 | **58.35** | 70.33 | 59.24 | **44.54** | 2.66 |
| Piatkowska adaptation [78] | **67.96** | 58.75 | **68.88** | **54.63** | 44.72 | 1.20 |
| Xie adaptation [105] | 93.43 | 78.60 | 93.66 | 66.57 | 59.36 | 1.39 |
| Kogler et al. [104] | 123.39 | 99.50 | 125.27 | 78.23 | 80.94 | 0.66 |
| Rogister et al. [103] | 142.17 | 128.20 | 153.32 | 114.80 | 90.32 | 1.01 |

MVSEC [35] indoor flying, raw outputs. Each method is scored on its own estimates and ground truth in $[0.5, 8]$ m; points summed and log RMSE averaged over sequences.

**Takeaway.** WLS is competitive with a larger scored population. This table does not compare accuracy on a common pixel set.

Dissertation: Table A.5, Table A.4; Xie and Piatkowska rows are local adaptations.

# A weighting comparison supports the selected rule

Compare reciprocal versus rate-proportional weights while holding the remaining configuration fixed.

| | Flying 1 | Flying 2 | Flying 3 | Flying 4 |
|---|---|---|---|---|
| Mean-error difference (cm) | 4.44 | 8.28 | 2.73 | 5.17 |

The reciprocal rule has lower reported mean error on all four sequences.

## Scope of this evidence

Each method is scored on its own valid pixel set. The differences subtract the reported WLS means from the WLS-Prop means; the rule was selected on these sequences.

**Takeaway.** The reported comparison supports the chosen rule on its scored populations; a common-set comparison requires rescoring the outputs.

Dissertation: Table A.5, Weighting comparison (p. 182).

# Hard fusion changes both the constraint and the regularization

Joint intensity fusion (JIF) imposes $\mathbf{f}_L = \mathbf{f}_R = \mathbf{v}$ and solves

$$(2\mathbf{A}^\top \mathbf{W}\mathbf{A} + \Lambda^2)\mathbf{v} = \mathbf{A}^\top \mathbf{W}(\mathbf{y}_L + \mathbf{y}_R).$$

For fixed equal weights, the unchanged soft model has the limit

$$\gamma \to \infty : \qquad \mathbf{f}_L, \mathbf{f}_R \to (\mathbf{A}^\top \mathbf{W}\mathbf{A} + \Lambda^2)^{-1} \frac{1}{2} \mathbf{A}^\top \mathbf{W}(\mathbf{y}_L + \mathbf{y}_R).$$

JIF instead uses $\mathbf{A}^\top \mathbf{W}\mathbf{A} + \frac{1}{2}\Lambda^2$ after dividing its equation by two.

**Takeaway.** Reported mean depth errors: JIF 102.10 cm; soft WLS 65.44 cm. Hard equality, regularization, weights, iteration budgets, and scored populations differ.

Dissertation: Proposition 4.9.1, Section 4.9.3.

# Motion-induced events and the aperture problem

For a straight edge under steady illumination, let $\mathbf{n}$ be its unit normal, $\mathbf{t}_\perp$ a unit tangent, and $s$ its normal speed:

$$L(\mathbf{x}, t) = \ell(\mathbf{n} \cdot \mathbf{x} - st), \qquad \nabla L = \ell' \mathbf{n}, \qquad L_t = -s\ell'.$$

Where $\ell' \neq 0$, brightness constancy gives

$$L_t + \nabla L \cdot \mathbf{v} = 0 \iff \mathbf{n} \cdot \mathbf{v} = s \iff \boxed{\mathbf{v} = s\mathbf{n} + \tau \mathbf{t}_\perp, \quad \tau \in \mathbb{R}.}$$

- Every such constant velocity gives the same local pattern and ideal events from the same reset reference.
- For $s = 0$, pixel values stay constant in the ideal edge interior; endpoints lie outside the observation window.

**Takeaway.** The edge determines normal speed; its tangential velocity remains free. Illumination changes can also trigger events under the sensor threshold rule.

Dissertation: Proposition 3.3.4 and following aperture remark; Eq. 2.1.

# One complementarity lemma behind the uniqueness results

**Lemma.** On a Hilbert space $H$, let $a, r \geq 0$ be bounded symmetric forms and $N \subset H$ finite-dimensional with orthogonal projection $P$. If

$$r(u,u) \geq \rho \|u - Pu\|^2, \qquad a(n,n) \geq \mu \|n\|^2 \quad (n \in N), \qquad a(u,u) \leq \beta \|u\|^2,$$

then $a(u,u) + r(u,u) \geq c_0 \|u\|^2$ with $c_0 = \min\{\mu/2,\ \rho\mu/(\mu + 2\beta)\}$. When $N = \ker r$, $\mu > 0$ is also necessary.

| Problem | $r$ controls all but $N$ | $N$ | $a$ must see $N$ |
|---|---|---|---|
| Temporal (Ch. 2) | data $\|\mathbf{Af}\|^2$, spectral gap $g_1$ | constants | penalty: $\sum_i \lambda_i^2 > 0$ |
| Flow (Ch. 3) | $\alpha^2 \|\nabla \mathbf{v}\|_{L^2}^2$, Poincaré | constant fields | data: $\lambda_{\min}(\int_\Omega \mathbf{B}) > 0$ |
| Discrete flow | connected graph Laplacian | constant fields | data: $\sum_i q_i B_i \succ 0$ |
| Stereo inner (Ch. 4) | data $+\gamma \|\mathbf{f}_L - \mathbf{f}_R\|^2$; $\mathbf{W}_c \succ 0$, $\gamma > 0$ | common constants | penalty: $\mathbf{\Lambda} \neq 0$ |

**Takeaway.** One term controls all but a finite-dimensional mode; the other must see it. The roles swap between temporal and flow, and $c_0$ falls linearly with $\mu$.

Dissertation: Theorem 2.2.5, Theorem 3.6.1, Theorem 3.7.1, Theorem 4.6.3. The common lemma restates their proofs.

# Adaptive weights keep the temporal estimate locally Lipschitz

Let $\lambda_i = \varphi(|dc_i|)$ with $0 < \lambda_- \le \varphi \le \lambda_+$ and $\varphi$ $L$-Lipschitz. Subtracting the normal equations,

$$\mathbf{K}_1(\mathbf{f}_1^* - \mathbf{f}_2^*) = \mathbf{A}^\top(\mathbf{y}_1 - \mathbf{y}_2) - (\mathbf{\Lambda}_1^2 - \mathbf{\Lambda}_2^2)\mathbf{f}_2^*.$$

With $\left\|\mathbf{K}^{-1}\mathbf{A}^\top\right\| \le \frac{1}{2\lambda_-}$, $\left\|\mathbf{K}^{-1}\right\| \le \lambda_-^{-2}$, $\left\|\mathbf{\Lambda}_1^2 - \mathbf{\Lambda}_2^2\right\| \le 2\lambda_+ L \left\|dc_1 - dc_2\right\|_\infty$:

$$\|\mathbf{f}_1^* - \mathbf{f}_2^*\| \le \frac{\|\mathbf{y}_1 - \mathbf{y}_2\|}{2\lambda_-} + \frac{\lambda_+ L}{\lambda_-^3}\, \|\mathbf{y}_2\|\, \left\|dc_1 - dc_2\right\|_\infty .$$

| Chapter 2 rule | $\lambda_-$ | $\lambda_+$ | $L$ |
|---|---|---|---|
| Sigmoid | $10^{-3}$ | 5.001 | 2.5 |
| Inverse proportional | $10^{-3}$ | 100 | 100 |
| Exponential decay | $10^{-3} + 10e^{-10}$ | 10.001 | 10 |

**Takeaway.** Each Chapter 2 rule gives a locally Lipschitz estimator with explicit, pessimistic constants; Chapter 4's solution-dependent IRLS weights need the fixed-point analysis.

Dissertation: Theorem 2.2.5; adaptive rules (p. 38). The Lipschitz bound is a short argument beyond the chapter.

# How the parameters were chosen

| Problem | Parameters | Selection |
|---|---|---|
| Temporal (Ch. 2) | Scale of each $\mathbf{\Lambda}$ rule; kernel and width | Grid sweep per rule; three rules compared by fidelity and CN |
| Flow (Ch. 3) | Smoothness weight $\alpha$ | Grid sweeps on MVSEC |
| Stereo (Ch. 4) | Coupling $\gamma$, latent count; weighting | Grid sweep; paired weighting comparison on the four sequences |

Antil–Sayre [2]: scalar weight $\lambda_2$. Chapter 2: adaptive diagonal $\mathbf{\Lambda}$, three rules evaluated.

## What the theorems guarantee

The results hold for every positive value: sweeps tune bias and accuracy; uniqueness and stability hold throughout.

## Principled selection rules

Discrepancy principle (Morozov 1966; Skellam noise level $\theta_+ + \theta_-$), L-curve (Hansen 1992), GCV (Golub–Heath–Wahba 1979), bilevel learning.

**Takeaway.** Sweeps chose the values; the theorems cover every positive choice.

Dissertation: Adaptive rules (p. 38); Table 2.7; flow sweeps (p. 164); Section 4.8; Definition 4.6.1. Parameter-choice rules: external works named above.

# Learned estimators face the same free directions

| Problem | Left free by the events | Model-based choice | Learned choice |
|---|---|---|---|
| Temporal | Constant offset at each pixel (Lemma 2.2.4) | Magnitude penalty $\mathbf{\Lambda}$ (Theorem 2.2.5) | Brightness learned from training images [12] |
| Flow | Motion along a straight edge (Figure 3.1) | $H^1$ smoothness, integrated data (Theorem 3.6.1) | Spatial context learned from data [39] |
| Stereo | Candidates with coincident right-view profiles (Proposition 4.6.20) | Spatial cost aggregation (SGM) | Learned matching cost |

## Combining them

Learn $\mathbf{\Lambda}$ in an outer problem [Antil–Di–Khatri 2020]; for $\mathbf{\Lambda} \succ 0$, $\mathbf{\Lambda} \mapsto \mathbf{f}^*(\mathbf{\Lambda})$ is single-valued and smooth.

## Data without a training set

ISS and ballistics: reconstructed from events alone, without reference images or training data.

**Takeaway.** The theorems separate what the events determine from what the regularizer supplies, in either form.

Dissertation: Lemma 2.2.4, Theorem 2.2.5, Theorem 3.6.1, Proposition 4.6.20; qualitative demonstrations (p. 44). External paper cited above.

# The spline regularizer is uniformly coercive under refinement

Cubic B-splines, uniform knot spacing $h$: $\Phi_j \geq 0$, $\sum_j \Phi_j \equiv 1$, $\int_\Omega \Phi_j \leq h^2$, and $u_h = \sum_j c_j \Phi_j$.

## Step 1: coefficients control the $L^2$ norm

$u_h^2 \leq \sum_j c_j^2 \Phi_j$ pointwise, so $\|u_h\|_{L^2}^2 \leq h^2 \|\mathbf{c}\|^2$.

## Step 2: both regularizer terms, for each flow component

$$\beta\, \mathbf{c}^\top \mathbf{K}_0 \mathbf{c} + \tau_{\text{ridge}} \|\mathbf{c}\|^2 \;\geq\; \beta \varepsilon_{\text{ell}} \|\nabla u_h\|_{L^2}^2 + \frac{2\varepsilon_{\text{ridge}}}{Mh^2} \|u_h\|_{L^2}^2 .$$

$Mh^2 \to |\Omega|$ (ratio 1.20 at $h = 8\,\text{px}$), so the $H^1$ coercivity constant stays bounded below as $h \to 0$; $\kappa_2(\mathbf{K}_{\text{reg}}) = O(h^{-2})$, the usual stiffness-plus-mass growth.

**Takeaway.** The ridge eigenvalue shrinks with refinement; the function-space penalty stays uniform.

Dissertation: Proposition 3.5.1; contrast-maximization parameters (p. 166). The uniform bound is a short argument beyond the chapter.

# Weak regularization: a limit in time, the aperture line in space

**Temporal**, $\mathbf{\Lambda} = \lambda \mathbf{D}$, fixed diagonal $\mathbf{D} \succ 0$

- Every $\lambda > 0$: $\mathbf{1}^\top \mathbf{D}^2 \mathbf{f}_\lambda = 0$, so the baseline rule is independent of $\lambda$.
- $\lambda \to 0$: $\mathbf{f}_\lambda \to \operatorname{argmin}\{\|\mathbf{Du}\| : \mathbf{Au} = \mathbf{y}\}$, an exact fit of the increments.
- $\mathbf{D} = \mathbf{I}$: noise gain $\left\|\mathbf{K}_\lambda^{-1}\mathbf{A}^\top\right\| \le \min\{\frac{1}{2\lambda}, \frac{1}{\sqrt{g_1}}\}$, with $\frac{1}{\sqrt{g_1}} \approx \frac{N_t}{\pi}$.

**Flow**, smoothness weight $\alpha^2$

- If $\mathbf{B} = 0$ on an open $U \subset \Omega$: for $\mathbf{v} \in H_0^1(U; \mathbb{R}^2)$, $\mathcal{B}(\mathbf{v}, \mathbf{v}) = \alpha^2 \|\nabla \mathbf{v}\|^2 \le \alpha^2 \|\mathbf{v}\|_{H^1}^2$, so the coercivity constant is at most $\alpha^2$.
- $\alpha = 0$, time surface: the minimizers are all $\mathbf{v}$ with $\nabla \mathcal{T} \cdot \mathbf{v} = 1$ wherever $\nabla \mathcal{T} \neq 0$, the aperture line at each pixel.

**Takeaway.** Without the temporal penalty the increments are fit exactly and noise accumulates along the chain; without spatial smoothing each pixel returns to its aperture line.

Dissertation: Regularized formulation (p. 16); Lemma 4.4.1; Proposition 4.6.23; Theorem 3.6.1; Proposition 3.3.4. The flow statements are a short argument beyond the chapter.

# Where there are no events, smoothness fills in harmonically

Let $U \subset \Omega$ be open, connected, and Lipschitz, with $\mathbf{B} = 0$ and $b = 0$ a.e. in $U$, and $\Gamma = \partial U \cap \Omega \neq \emptyset$.

## The weak equation inside $U$

For every $\boldsymbol{\eta} \in H^1_0(U; \mathbb{R}^2)$: $\int_U \nabla \mathbf{v}^* : \nabla \boldsymbol{\eta} = 0$. Each component of $\mathbf{v}^*$ is harmonic, hence smooth, in $U$; on $U$ it minimizes $\int_U |\nabla \mathbf{w}|^2$ given its values on $\Gamma$.

## Maximum principle

For a unit vector $\mathbf{e}$: if $\mathbf{e} \cdot \mathbf{v}^* \leq M$ on $\Gamma$, test with $\boldsymbol{\eta} = (\mathbf{e} \cdot \mathbf{v}^* - M)^+ \mathbf{e}$ on $U$, zero elsewhere. Then $\alpha^2 \int_U |\nabla \boldsymbol{\eta}|^2 = 0$, so $\mathbf{e} \cdot \mathbf{v}^* \leq M$ in $U$. Over all $\mathbf{e}$: $\sup_U |\mathbf{v}^*| \leq \sup_\Gamma |\mathbf{v}^*|$.

**Takeaway.** An event-free region receives the least-gradient interpolation of the motion measured around it, with no speed larger than on its boundary.

Dissertation: Theorem 3.3.9; Theorem 3.6.1. Harmonicity and the maximum principle are a short argument beyond the chapter.

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